Heart Attack Prediction using ML Approach

By

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- Subject – What is heart attack?
  - How to interpret heart attack biologically? Symptoms of heart attack.
- Let's know the Dataset - Understanding features/variables
- Pipeline – what are we planning to do?
- Preprocessing- Missing values, Unique values, Variable categories
- EDA – Univariate Analysis, Bivariate Analysis
- Modelling preparation
- Machine Learning Inference analysis

Domain – Subject

- What is heart attack?
  - Medical Emergency
  - Chest pain
  - Shortness of breath
  - Unstoppable cold sweatness

- Interpret heart attack biologically
  - Blockage in coronary artery
  - Blood clot in Coronary Artery
  - Narrowing of Coronary Artery

- Symptoms of heart attack

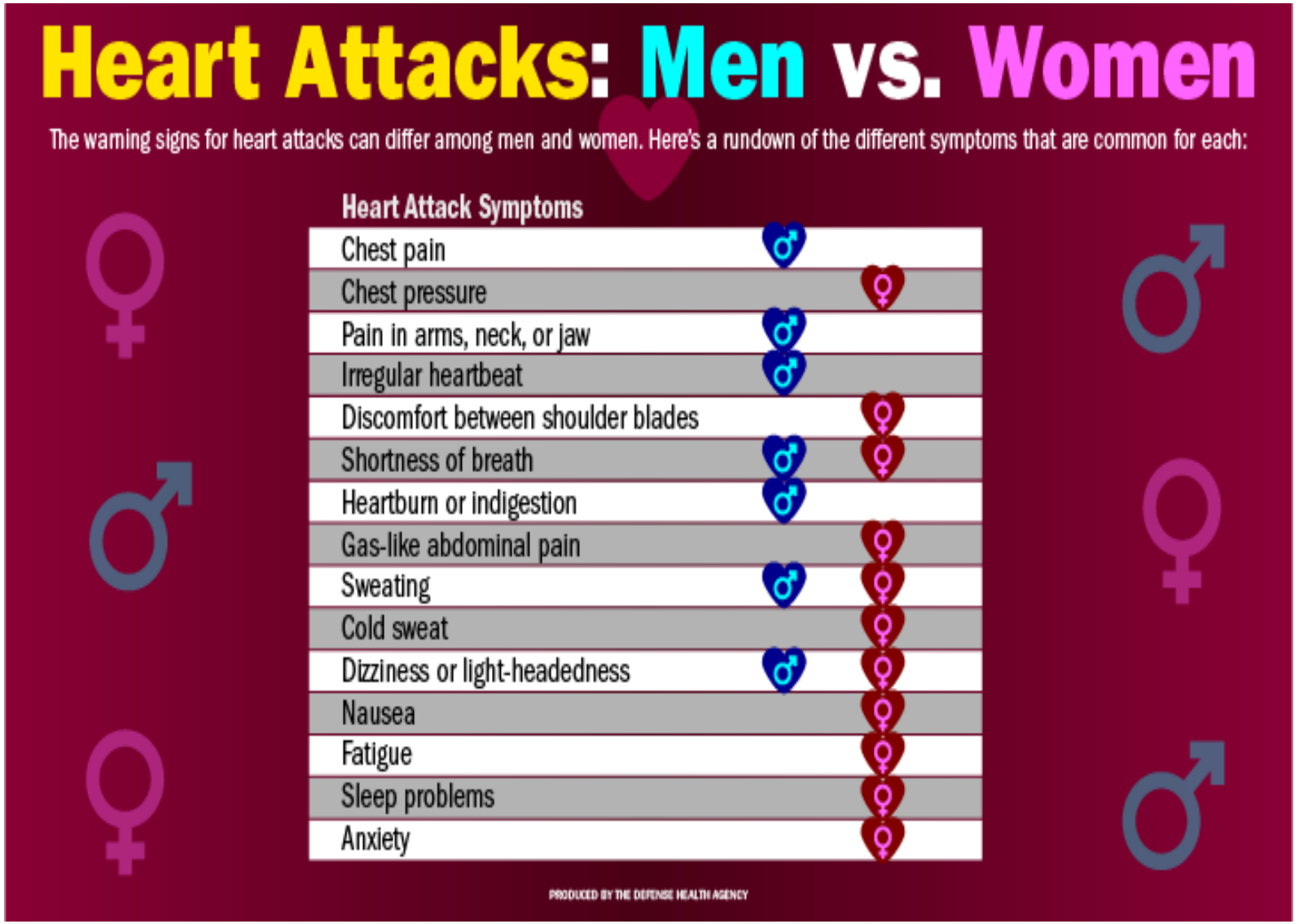


Image Source : https://health.mil/News/Articles/2022/02/11/Womens-Heart-Attacks-Symptoms-Can-Differ-from-Mens-Know-the-Signs?type=Photos

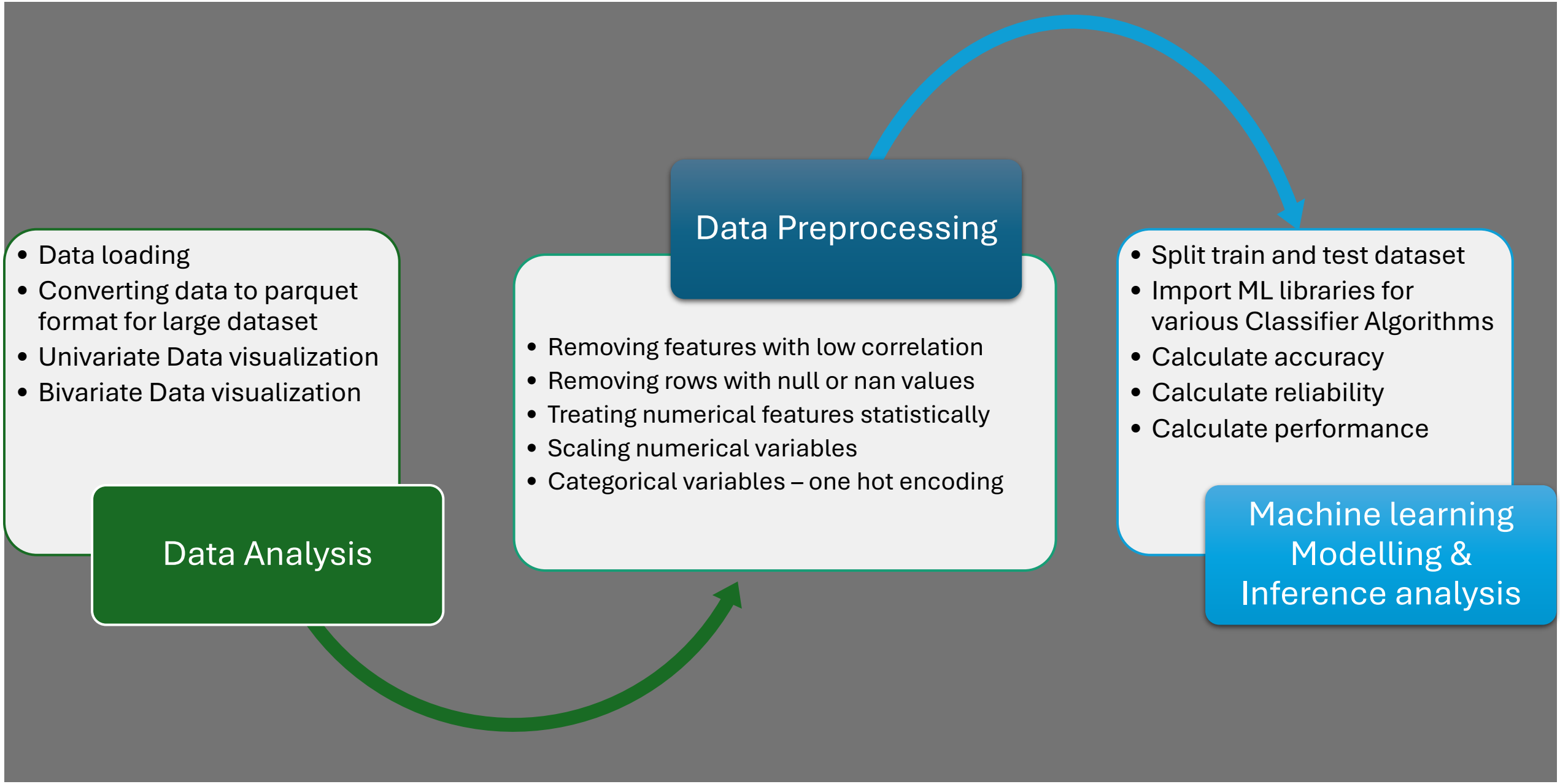
Dataset - Understanding features/variables

Variable Name	#	Definition	Values
1. #3 (age)	1	**Age:** The age of the individual.	age: age in years
2. #4 (sex)	2	**Sex: **The gender of the individual.	(1 = male; 0 = female)
4. #10 (trestbps)	4	**Resting Blood Pressure:** The resting blood pressure of the individual (in mm Hg).	trestbps: resting blood pressure (in mm Hg on admission to the hospital)
5. #12 (chol)	5	**Cholesterol: **The cholesterol levels of the individual (in mg/dL).	chol: serum cholestoral in mg/dl
6. #16 (fbs)	6	**Fasting Blood Sugar: **The fasting blood sugar level of the individual (> 120 mg/dL is considered high).	(fasting blood sugar > 120 mg/dl) (1 = true; 0 = false)
7. #19 (restecg)	7	**Resting Electrocardiographic Results (ECG): **Results of the resting electrocardiogram.	restecg: resting electrocardiographic results -- Value 0: normal -- Value 1: having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV) -- Value 2: showing probable or definite left ventricular hypertrophy by Estes' criteria
10. #40 (oldpeak)	10	**ST Depression Induced by Exercise: **The ST depression induced by exercise relative to rest.	oldpeak = ST depression induced by exercise relative to rest < 1mm
11. #41 (slope)	11	**Slope of the Peak Exercise ST Segment: **The slope of the peak exercise ST segment.	slope: the slope of the peak exercise ST segment -- Value 1: upsloping -- Value 2: flat -- Value 3: downsloping
12. #44 (ca)	12	**Number of Major Vessels Colored by Fluoroscopy: **The number of major blood vessels colored by fluoroscopy.	ca: number of major vessels (0-3) colored by flourosopy
13. #51 (thal)	13	**Thalassemia: **A blood disorder; different types of thalassemia might be represented in the dataset.	Thalium stress test: thal: 3 = normal; 6 = fixed defect; 7 = reversable defect
14. #58 (num) (the predicted attribute)	14	**Target: **Whether the individual had a heart attack or not (typically represented as binary: 0 for no heart attack, 1 for heart attack).	diagnosis of heart disease (angiographic disease status) -- Value 0: < 50% diameter narrowing > Less chance of Heart attack -- Value 1: > 50% diameter narrowing > More chance of Heart attack

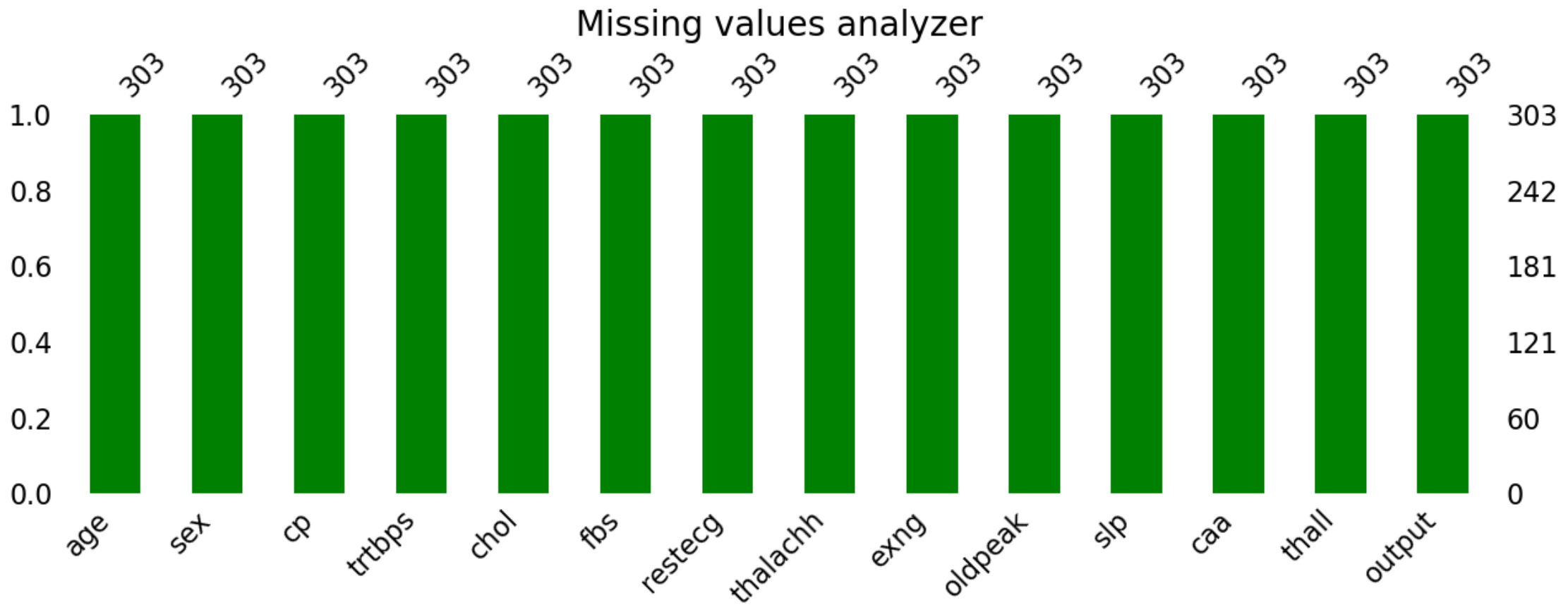
Dataset - Understanding features/variables

Variable Name	#	Definition	Values
3. #9 (cp)	3	**Chest Pain Type:** The type of chest pain experienced by the individual.	cp: chest pain type -- Value 1: typical angina - **Chest Pain occurred during physical activity.** Classic Pain. -- Value 2: atypical angina - Pain due to shortness of breath. -- Value 3: non-anginal pain - Pain that is caused by non cardiac or heart disease. -- Value 0: asymptomatic - Pain that occurs without symptoms
8. #32 (thalach)	8	**Maximum Heart Rate Achieved:** The maximum heart rate achieved by the individual during exercise.	thalach: maximum heart rate achieved. Normal range 60 to 100 beats per minute
9. #38 (exang)	9	**Exercise Induced Angina. Angina - is chest pain or discomfort caused by reduced blood flow to the heart muscle - Involuntary contraction of the coronary artery. **Whether the individual experienced angina during exercise.	exang: exercise induced angina (1 = yes; 0 = no)

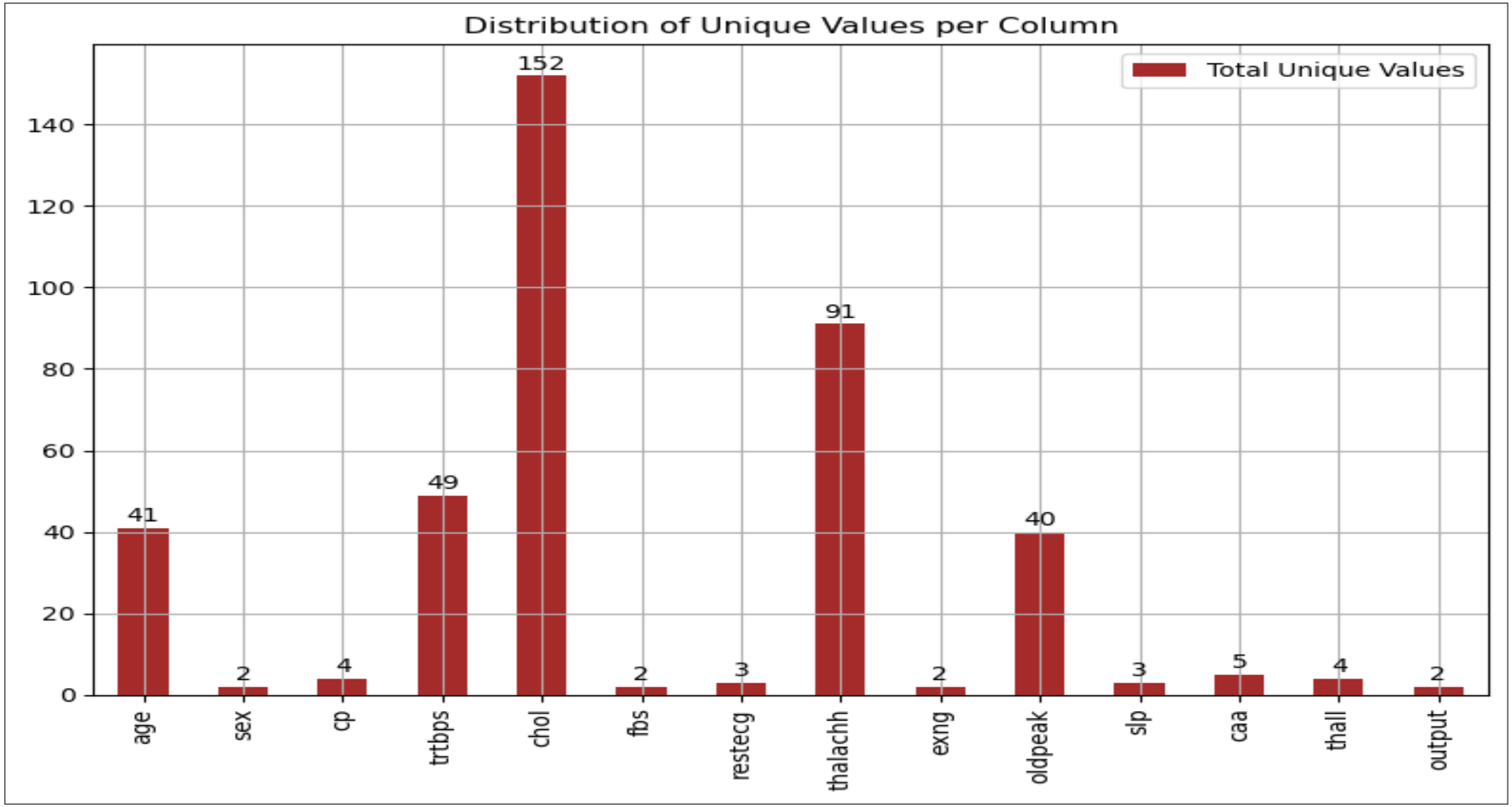
Machine Learning Pipeline



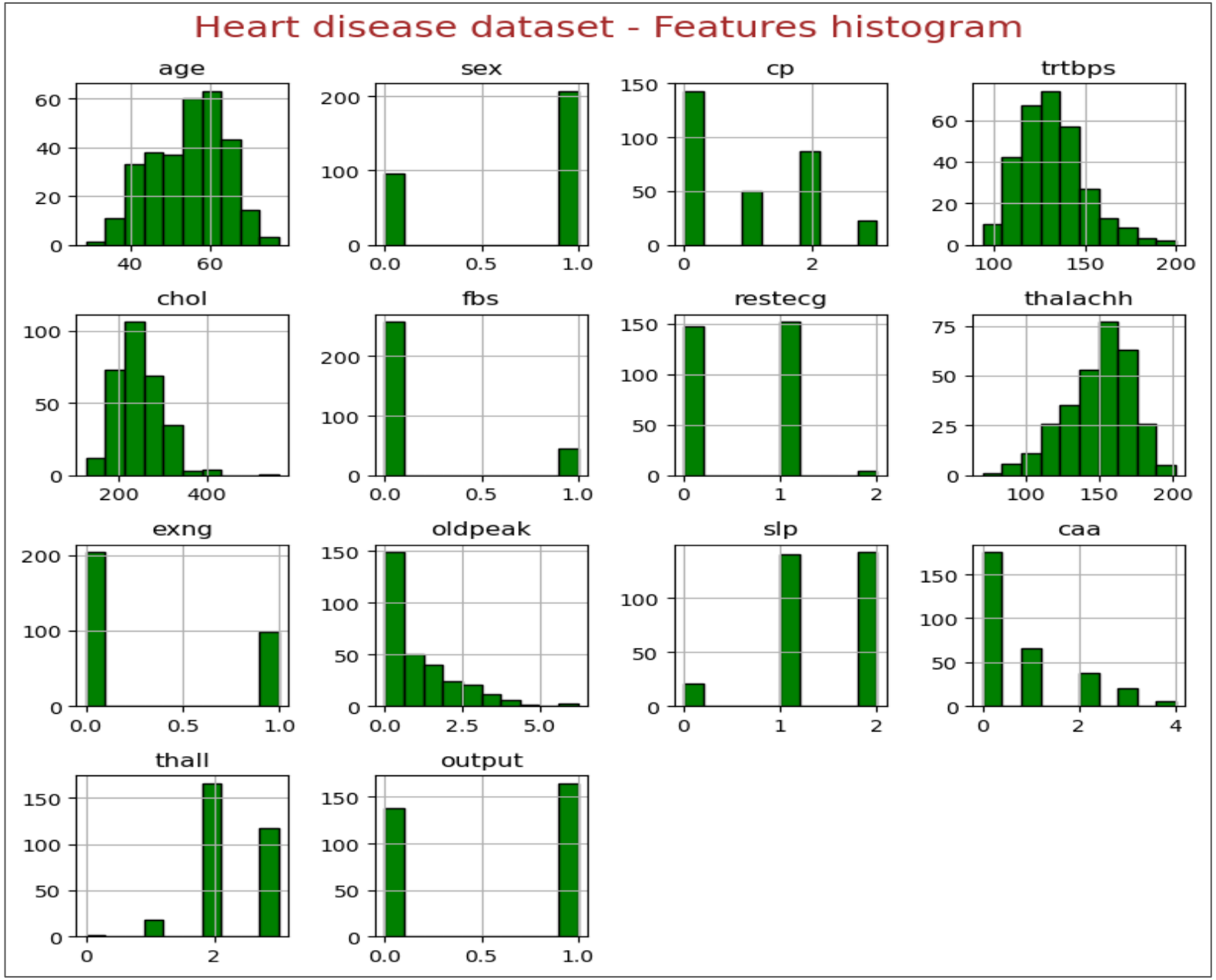
Preprocessing- Missing values – missingno package

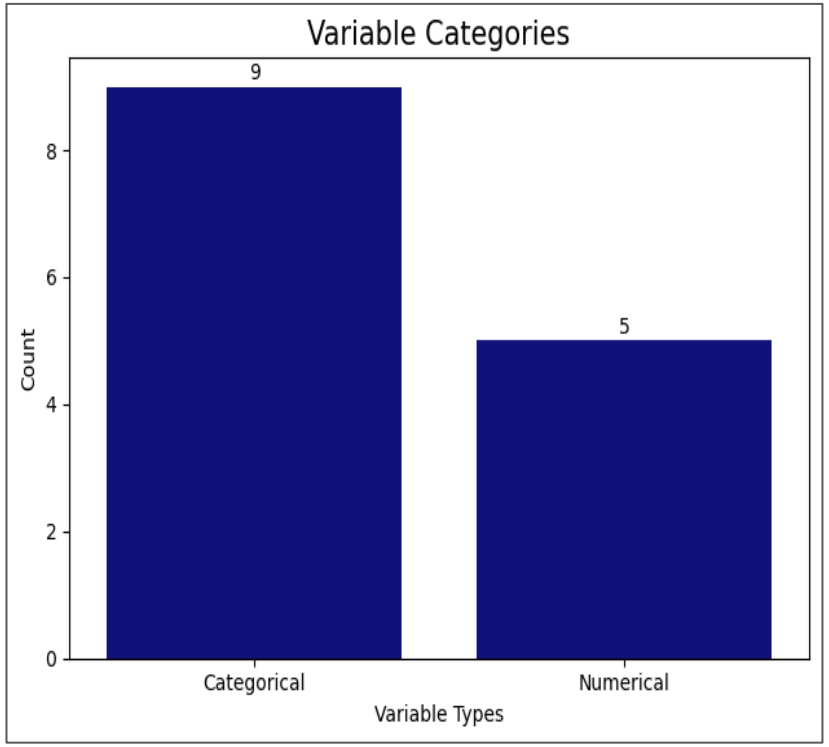


Preprocessing- Unique values



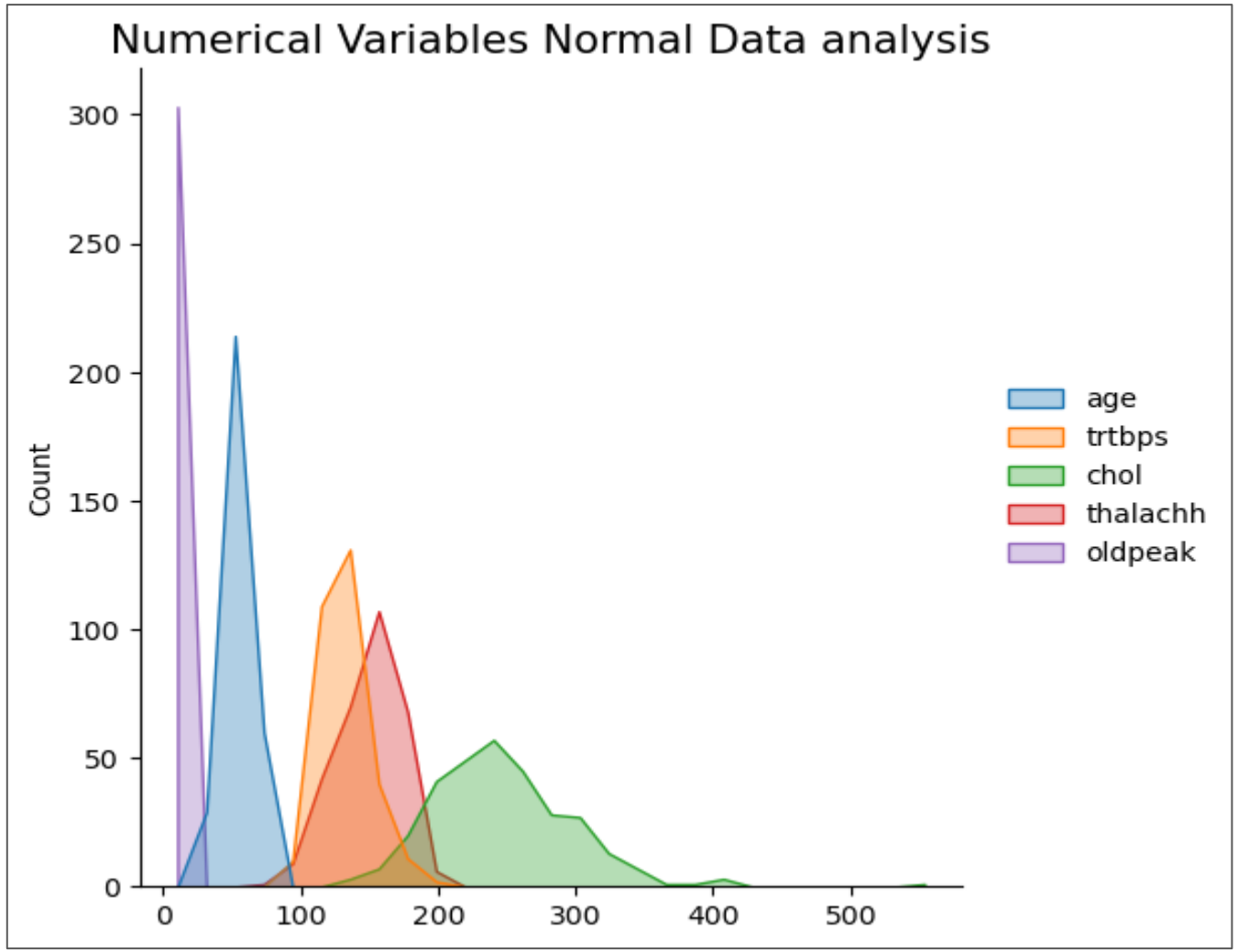
Preprocessing- variable categories





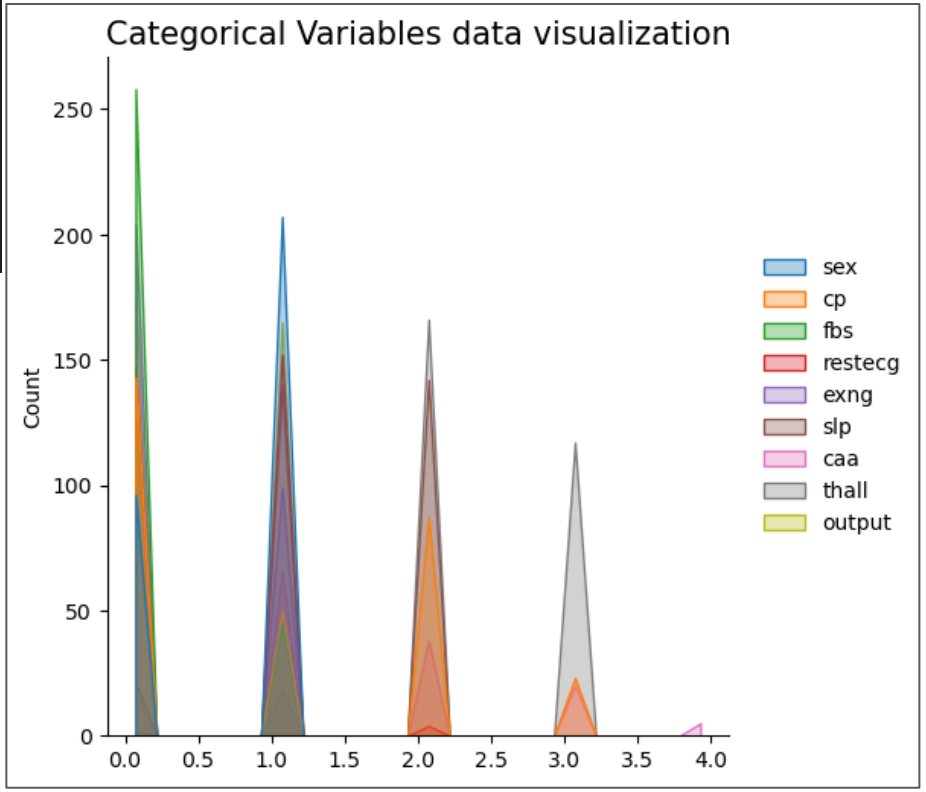
Preprocessing- Examining Statistics of Numerical variables

	age	trtbps	chol	thalachh	oldpeak
count	303	303	303	303	303
mean	54.3663	131.624	246.264	149.647	1.0396
std	9.0821	17.5381	51.8308	22.9052	1.16108
min	29	94	126	71	0
25%	47.5	120	211	133.5	0
50%	55	130	240	153	0.8
75%	61	140	274.5	166	1.6
max	77	200	564	202	6.2

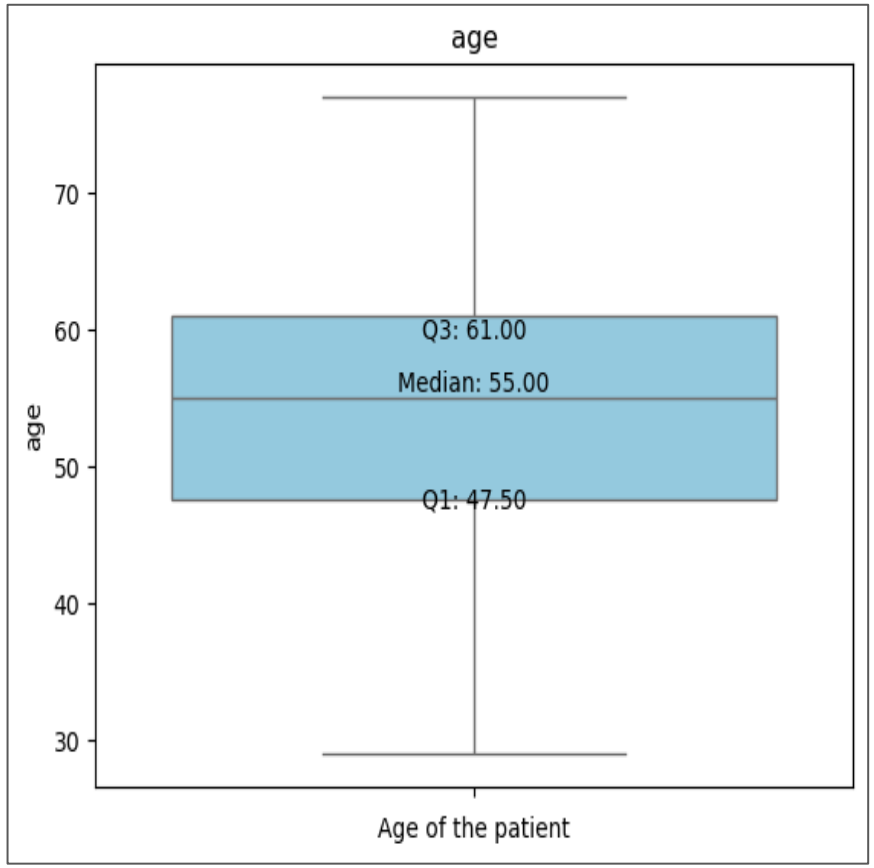


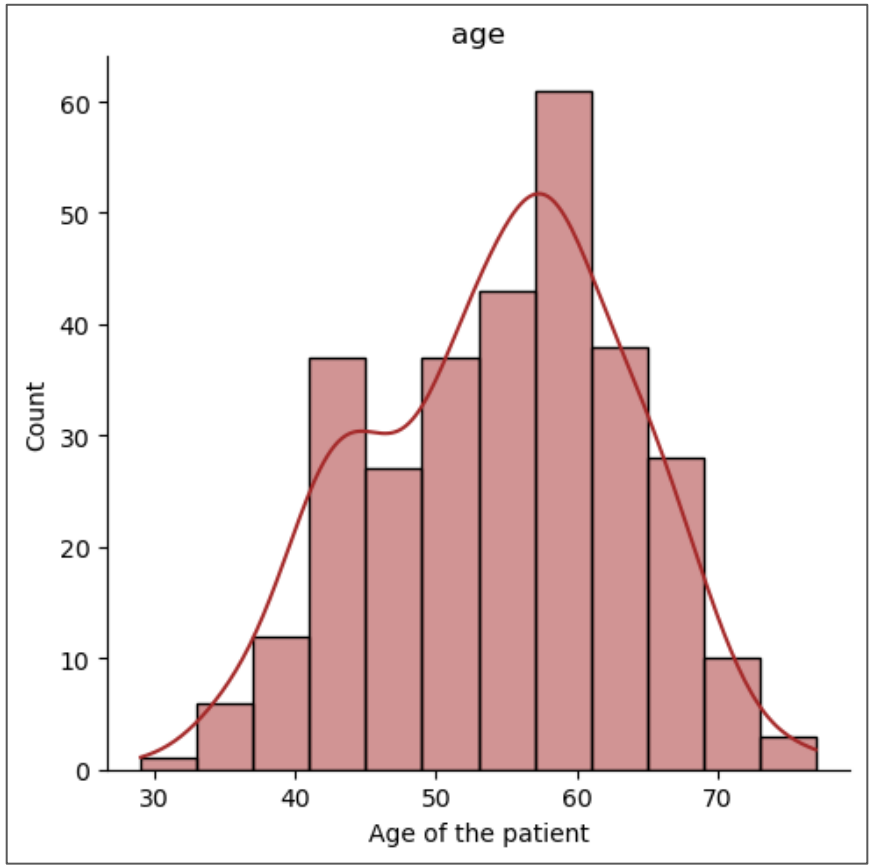
Preprocessing- Examining Statistics of Categorical variables

	sex	cp	fbs	restecg	exng	slp	caa	thall	output
count	303	303	303	303	303	303	303	303	303
mean	0.683168	0.966997	0.148515	0.528053	0.326733	1.39934	0.729373	2.31353	0.544554
std	0.466011	1.03205	0.356198	0.52586	0.469794	0.616226	1.02261	0.612277	0.498835
min	0	0	0	0	0	0	0	0	0
25%	0	0	0	0	0	1	0	2	0
50%	1	1	0	1	0	1	0	2	1
75%	1	2	0	1	1	2	1	3	1
max	1	3	1	2	1	2	4	3	1



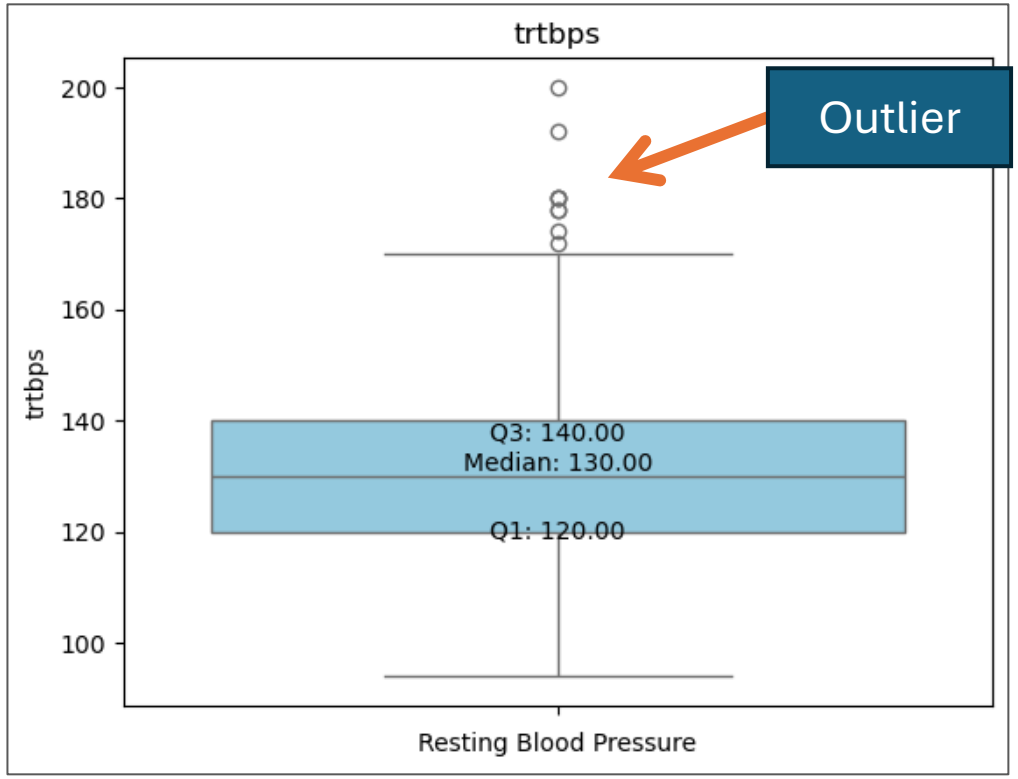
EDA – Univariate Analysis – Boxplot and Histplot - Age

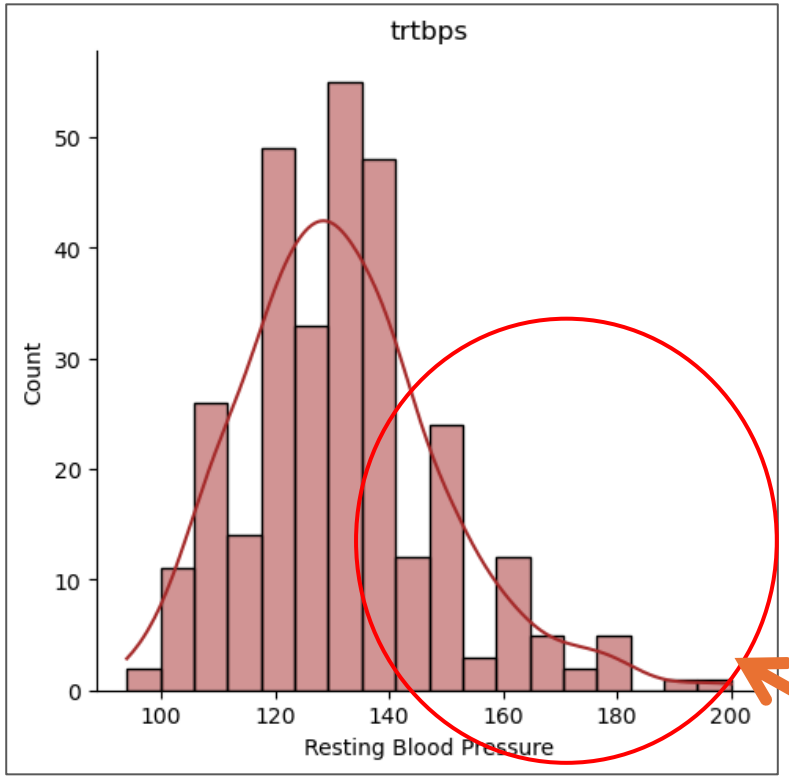




Analysis – No significant observation. Normal age expectancy

EDA – Univariate Analysis – Boxplot and Histplot - BP

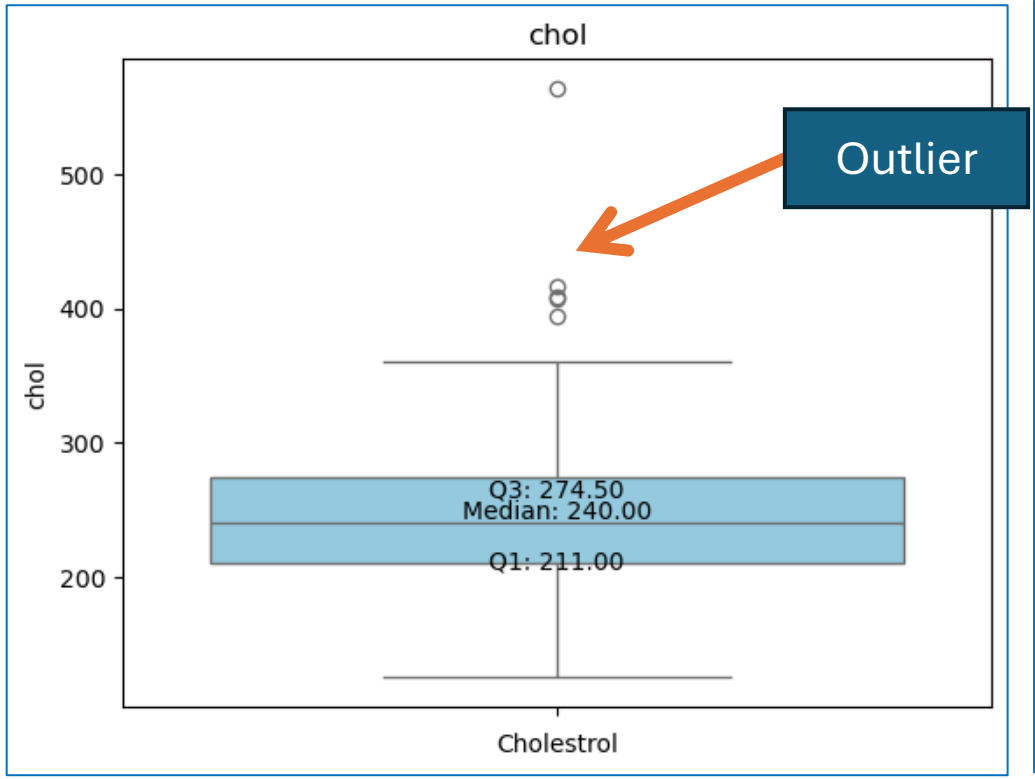




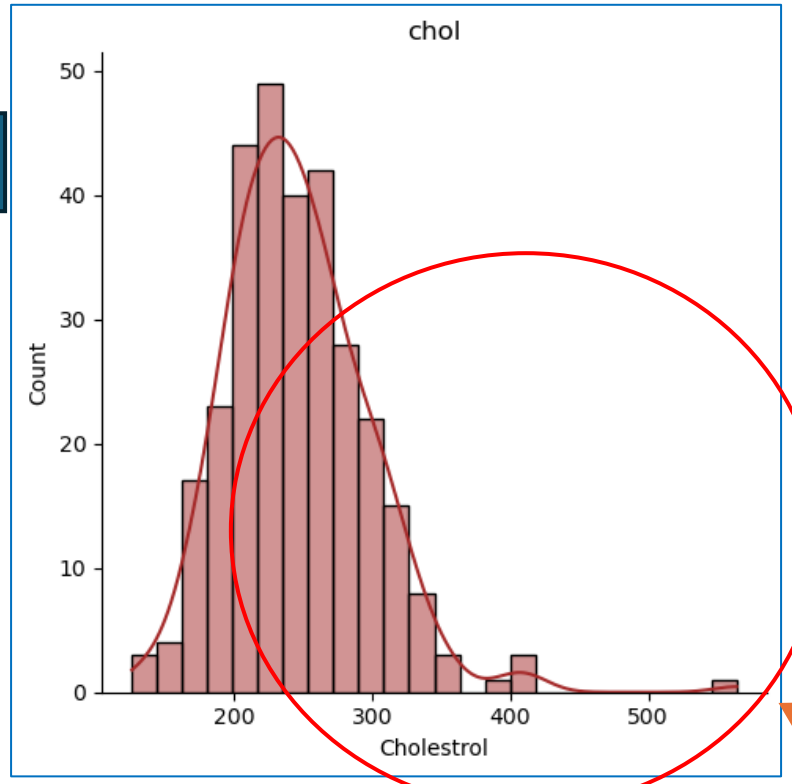
Analysis – outlier exists. Normal range – 80-120mmHg

Outlier

EDA – Univariate Analysis – Boxplot and Histplot - Cholesterol



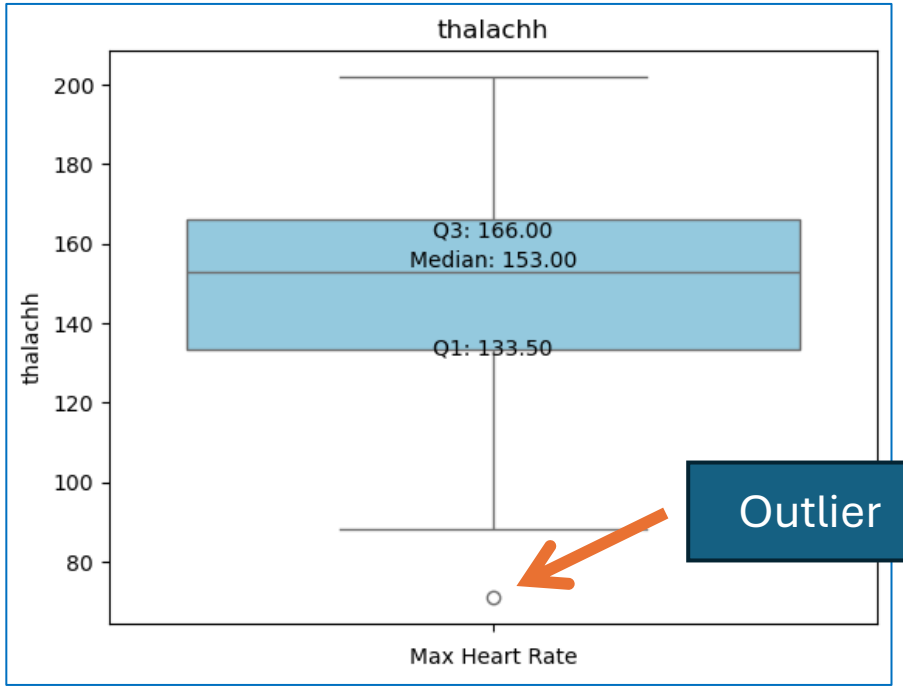
Outlier



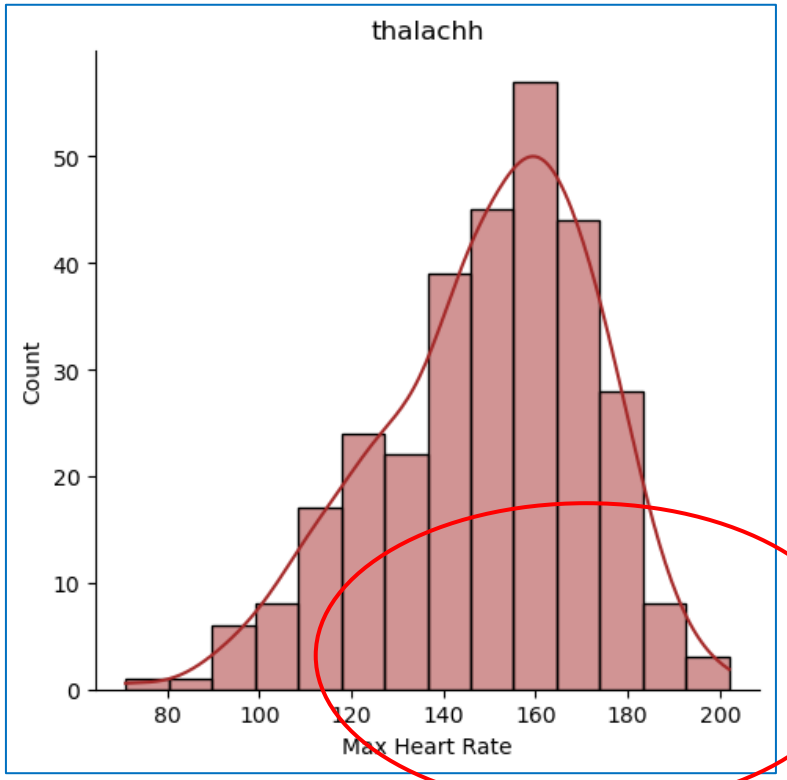
Analysis – outlier exists. Normal range – less than 200

Outlier

EDA – Univariate Analysis – Boxplot and Histplot – Thalach - HR



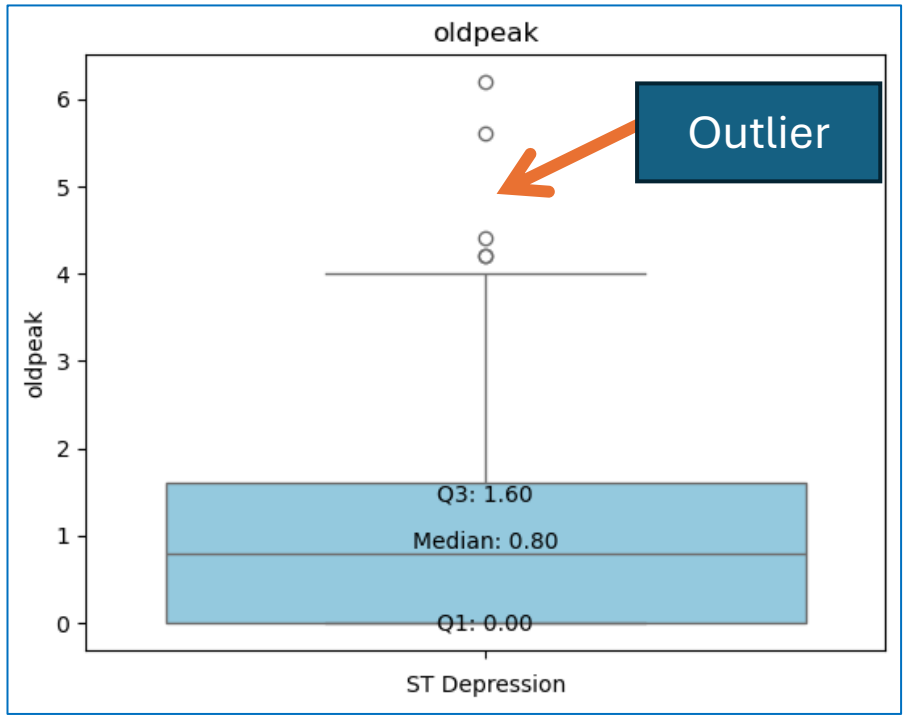
Outlier

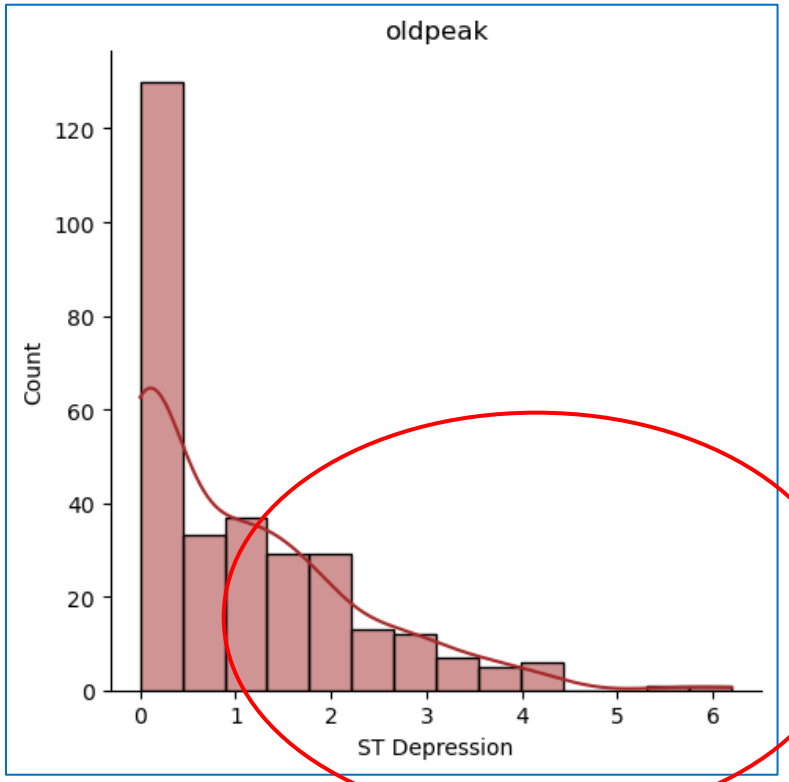


Analysis – outlier exists. Normal range 60 to 100 beats per minute

Outlier

EDA – Univariate Analysis – Boxplot and Histplot – ST Depression

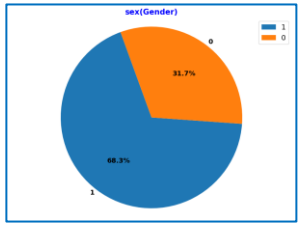


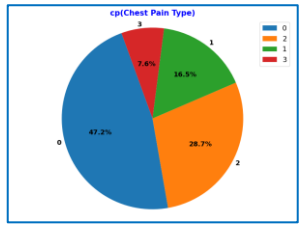


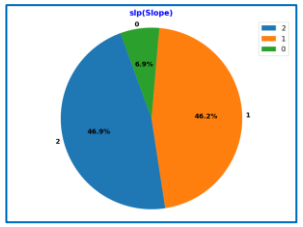
Analysis – outlier exists. Normal range less than 1 mm

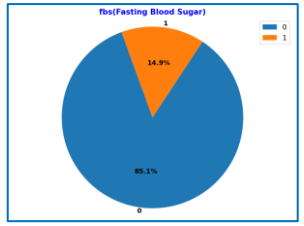
Outlier

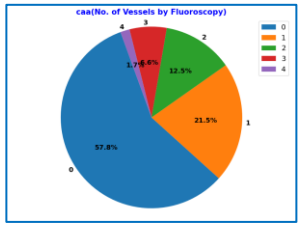
EDA – Univariate Analysis – pie chart for categorical features

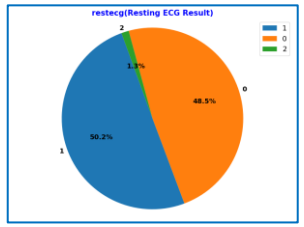


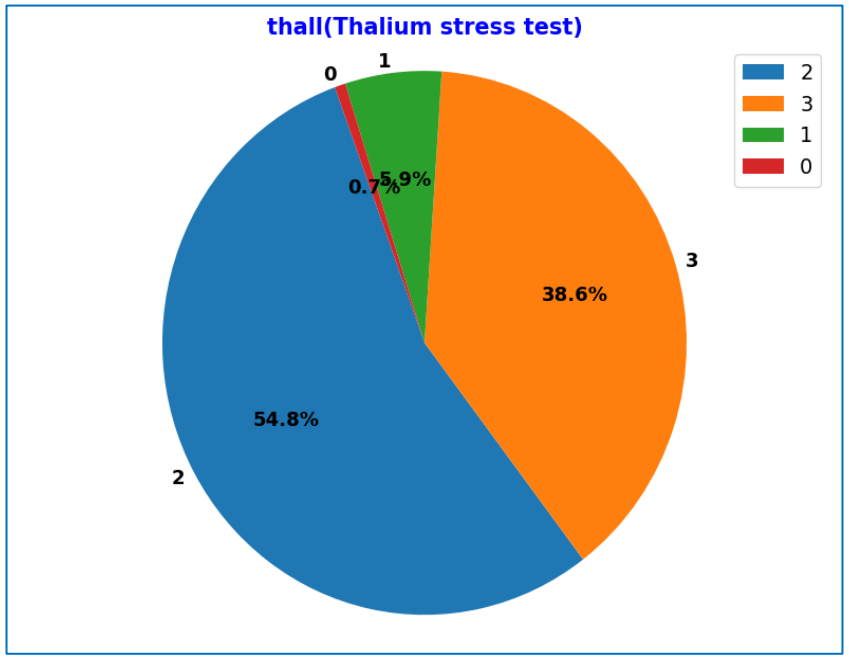




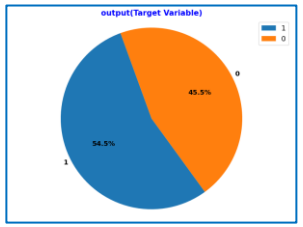


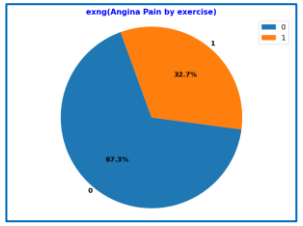






Analysis – outlier exists.
Thal – Thalium stress test has value 0 for 3
0 is Outler and is replaced with mas of the column.



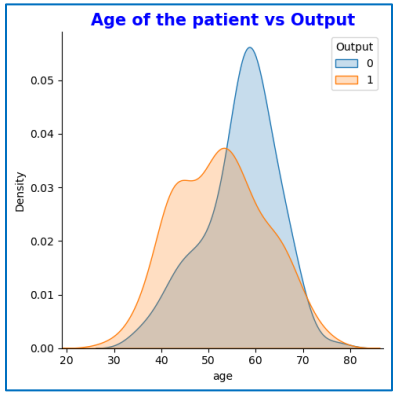


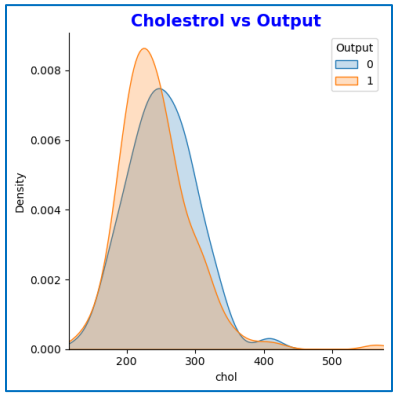
	age	sex	cp	trtbps	chol	fbs	restecg	thalachh	exng	oldpeak	slp	caa	thall	output
48	53	0	2	128	216	0	0	115	0	0	2	0	0	1
281	52	1	0	128	204	1	1	156	1	1	1	0	0	0

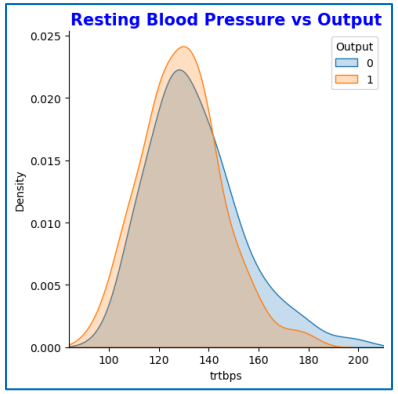
Replacing 0 with max of thal

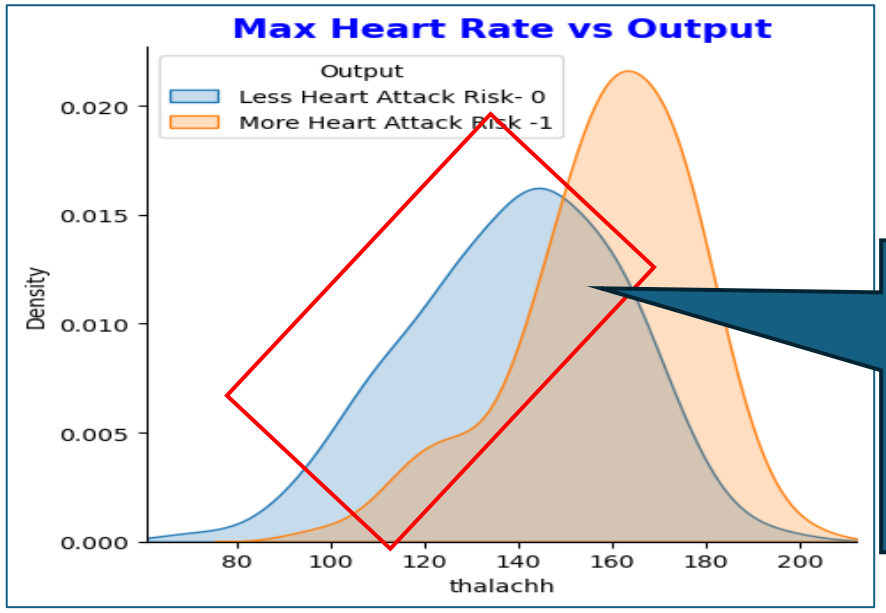
	age	sex	cp	trtbps	chol	fbs	restecg	thalachh	exng	oldpeak	slp	caa	thall	output
48	53	0	2	128	216	0	0	115	0	0	2	0	3	1
281	52	1	0	128	204	1	1	156	1	1	1	0	3	0

EDA – Bivariate Analysis – Numerical Features vs Output

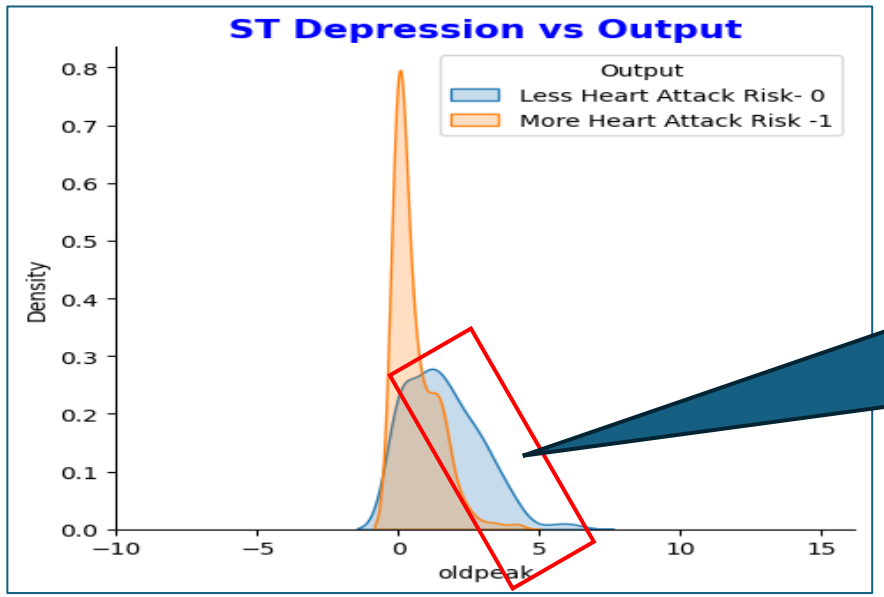








Positive Correlation wrt Output



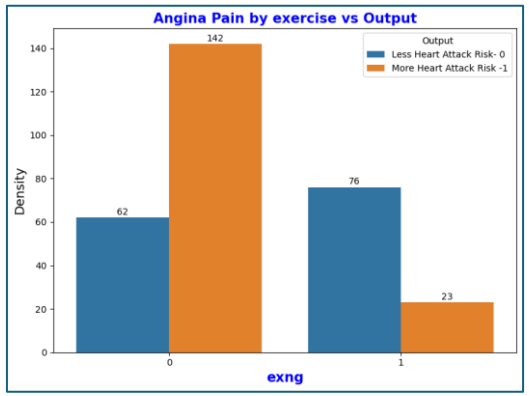
Negative Correlation wrt Output

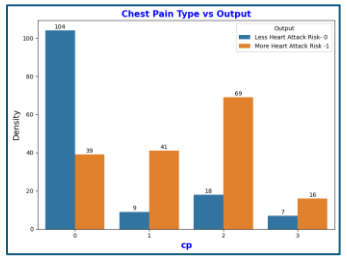
Features don't stand separate with output. Hence most of the features are partially correlated

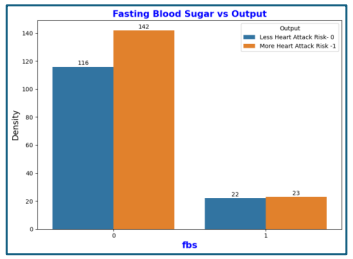
```
print(ha_df[numeric_var_uvdef].corr().iloc[:, [-1]])
```

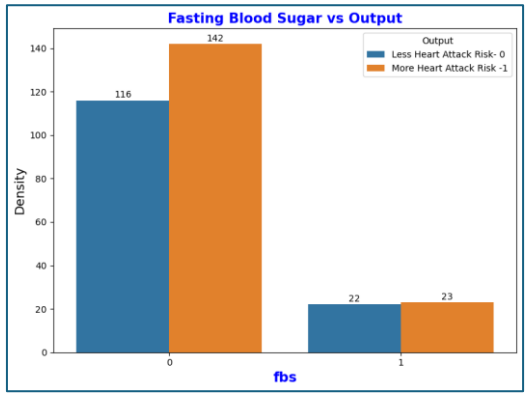
	output
age	-0.225439
trtbps	-0.144931
chol	-0.085239
thalachh	0.421741
oldpeak	-0.430696
output	1.000000

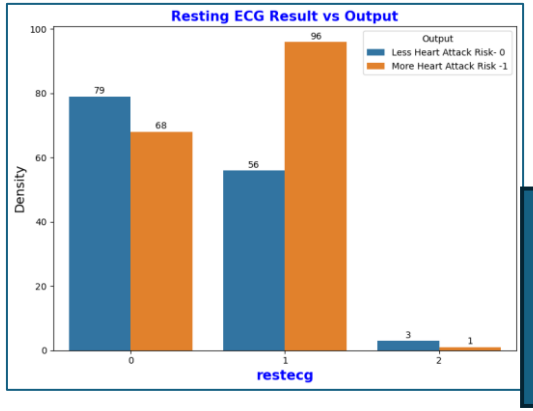
EDA – Bivariate Analysis – Categorical Feature vs Output





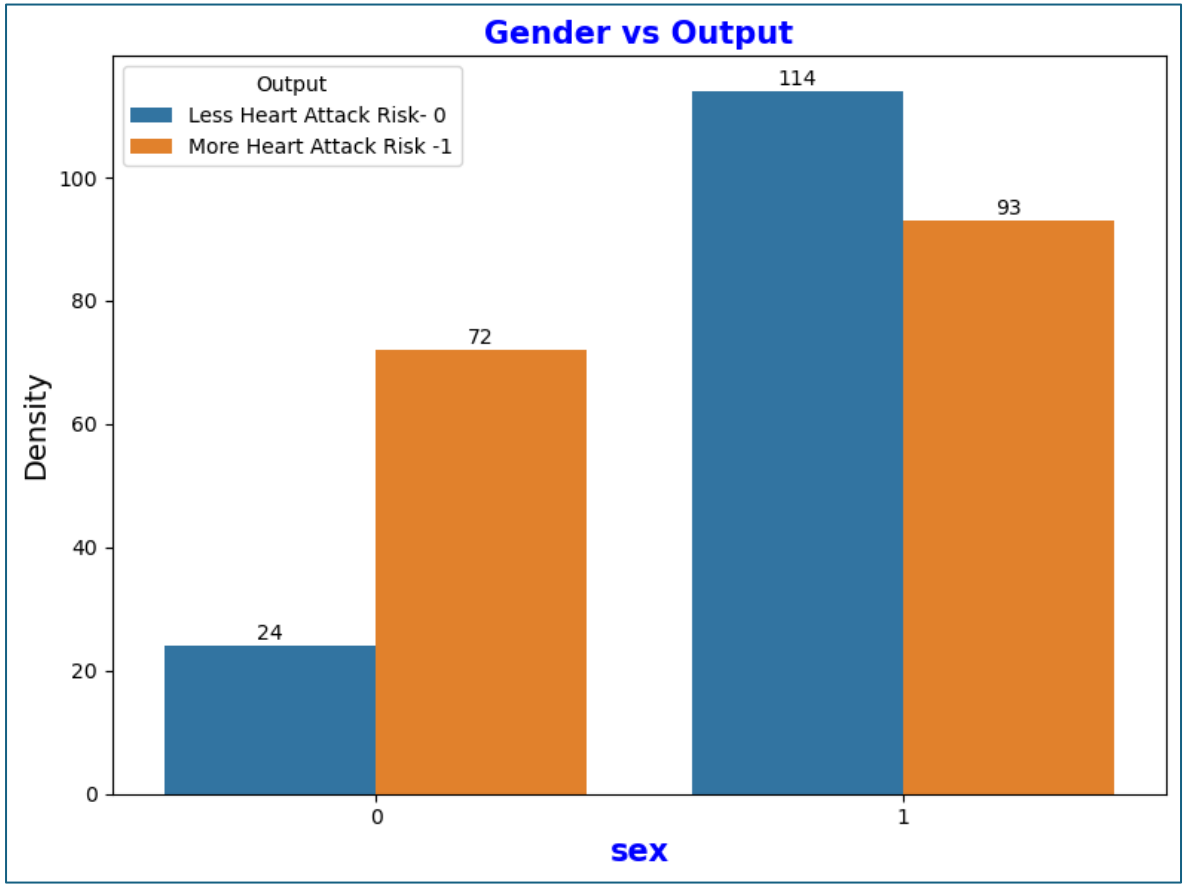




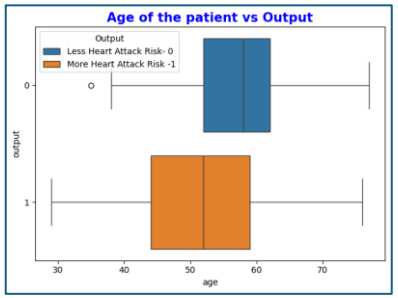


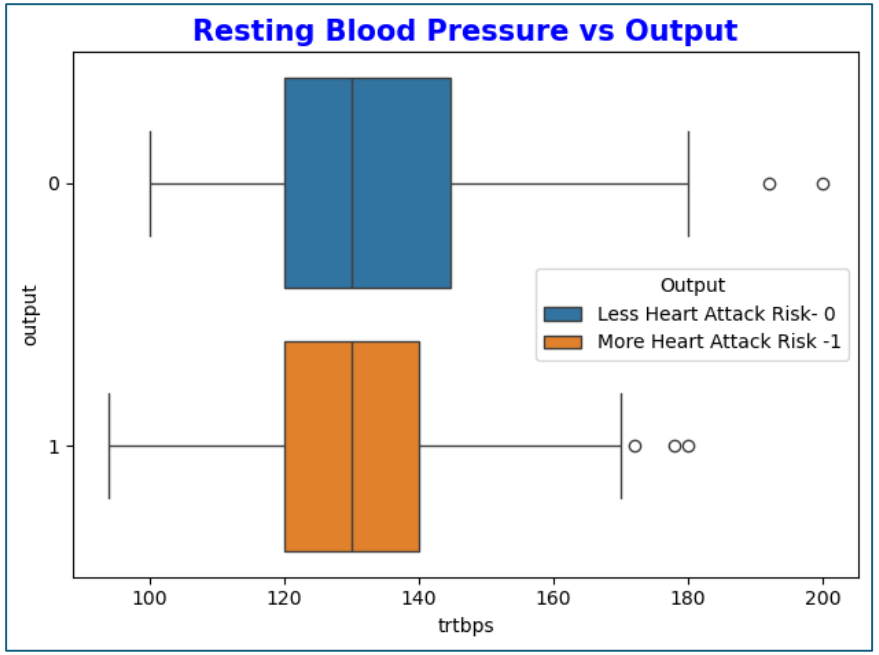
	output
sex	-0.280937
cp	0.433798
fbs	-0.028046
restecg	0.137230
exng	-0.436757
slp	0.345877
caa	-0.391724
thall	-0.363136
output	1.000000

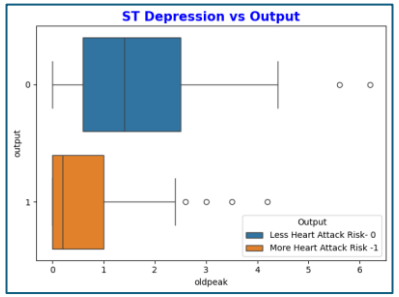
Negative Correlation wrt Output

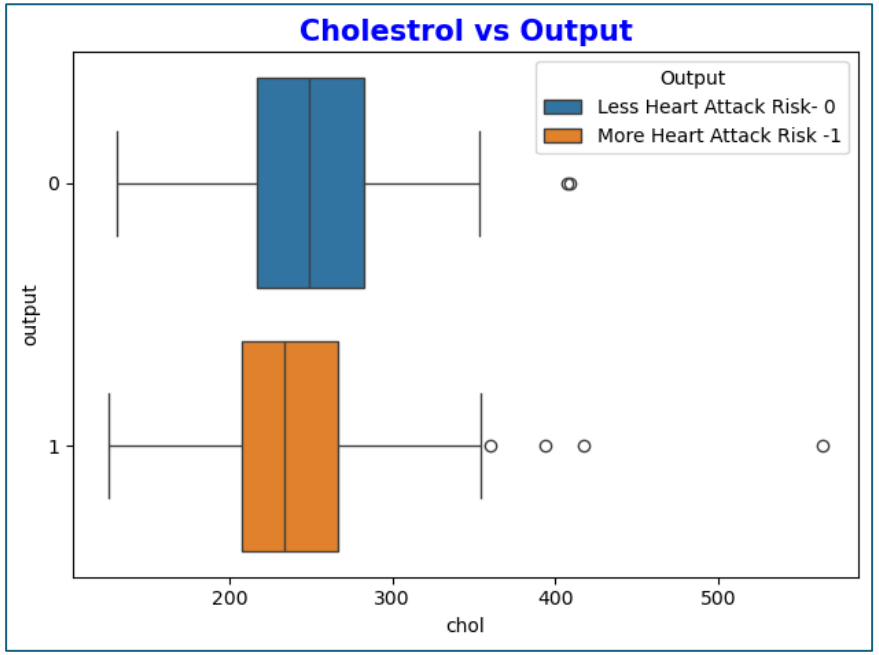


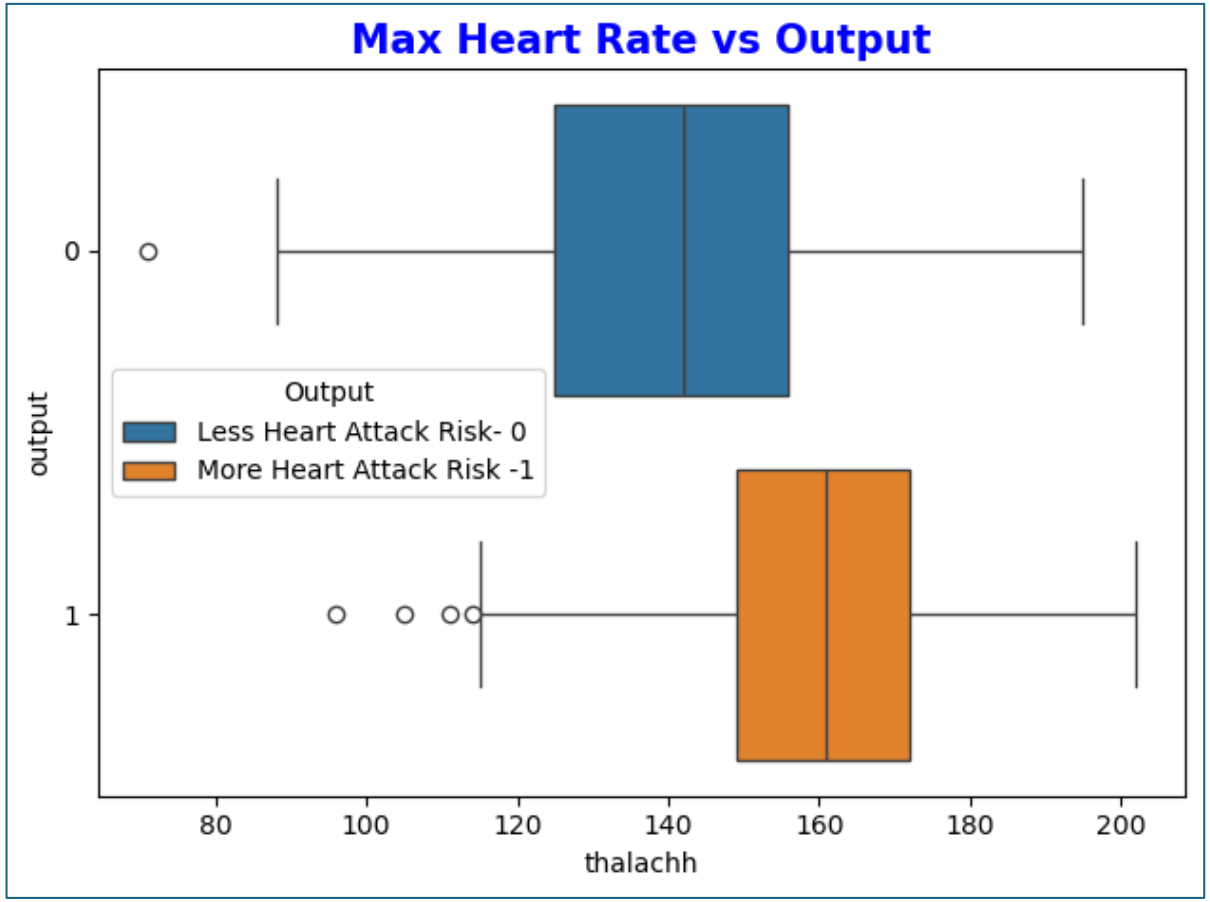
EDA – Bivariate Analysis – Box Plot for numerical variables





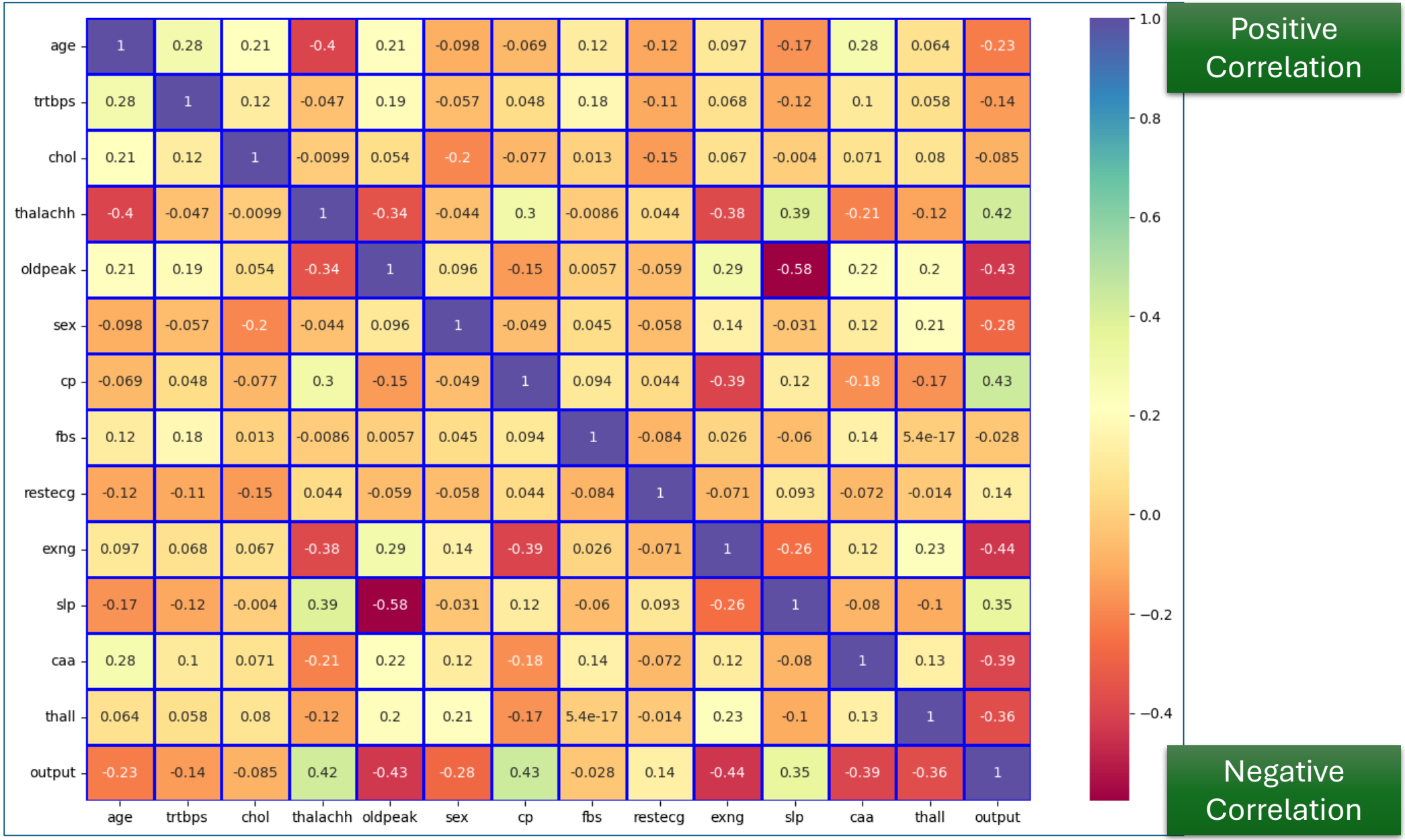






Preparation for modelling

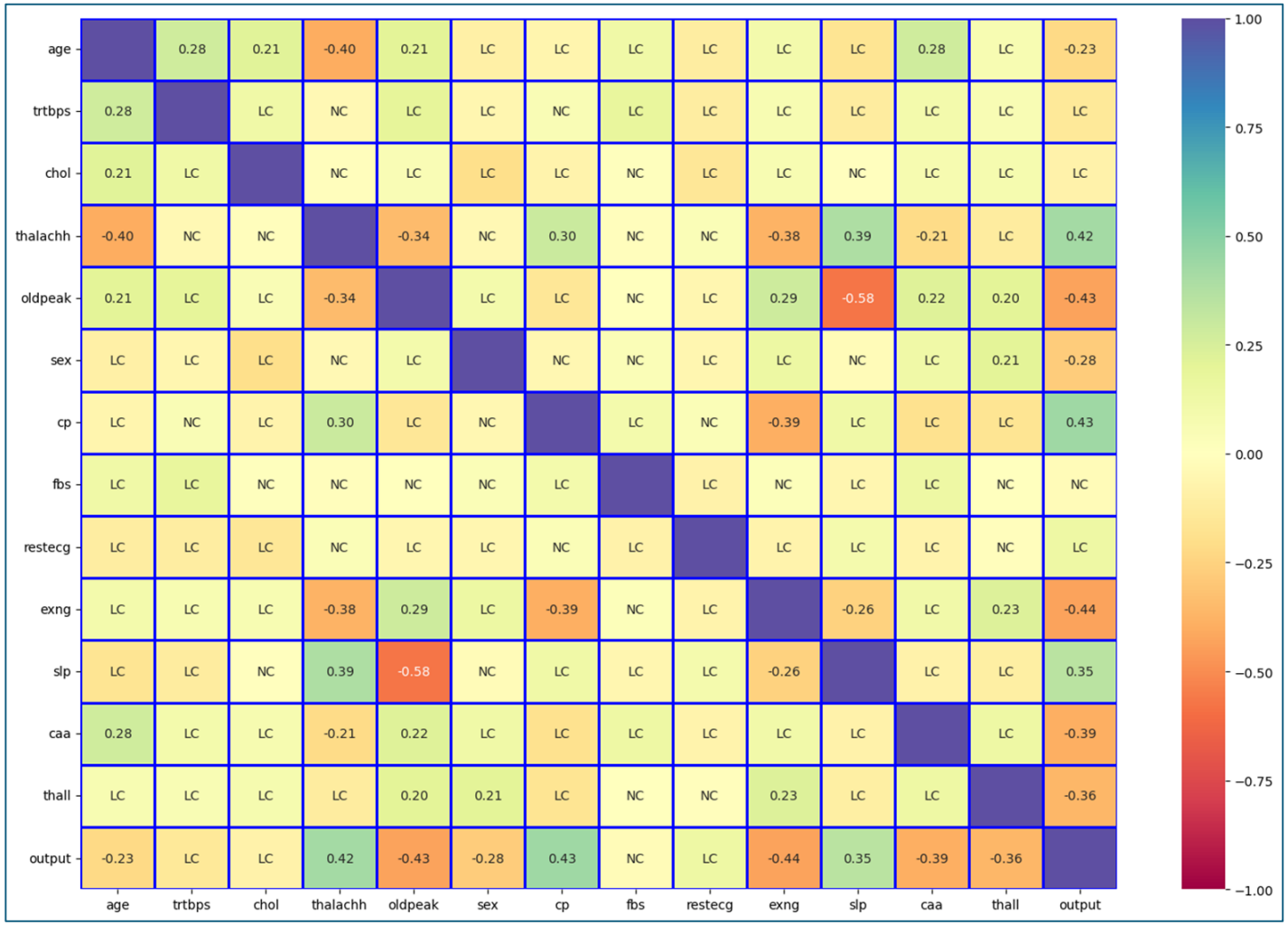
Heat Map – Removing low correlation variables



Positive Correlation

Negative Correlation

EDA – Heat Map – Removing low correlation variables



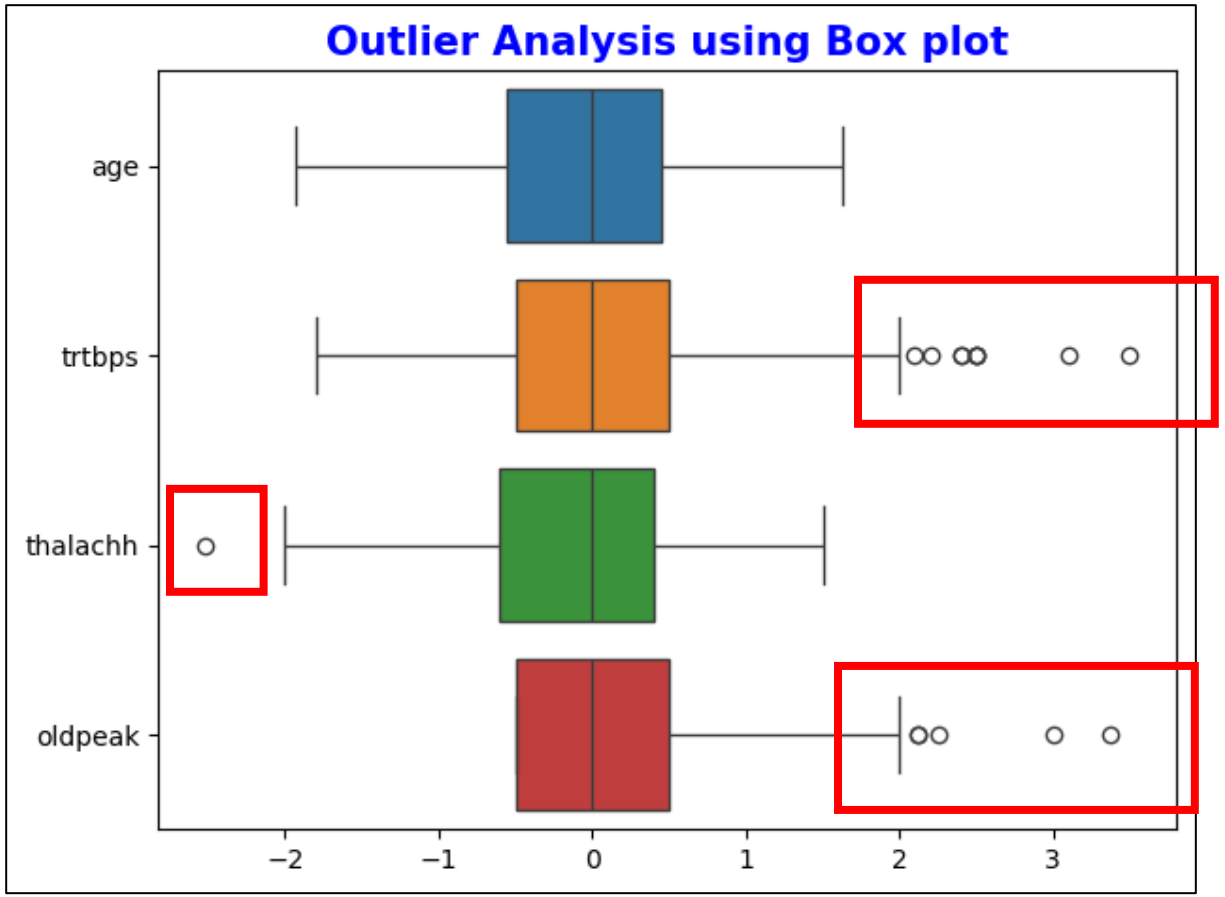
Removing Chol, fbs and restecg as they are either low or no correlation with respect to output variable



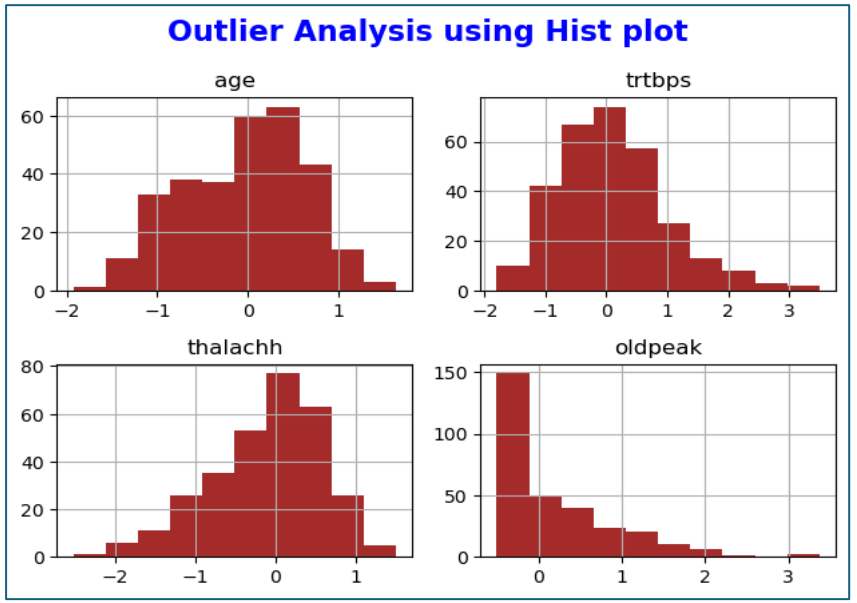
Updated Dataframe

	age	trtbps	thalachh	oldpeak	sex	cp	exng	slp	caa	thall	output
0	0.592593	0.75	-0.0923077	0.9375	1	3	0	0	0	1	1
1	-1.33333	0	1.04615	1.6875	1	2	0	0	0	2	1
2	-1.03704	0	0.584615	0.375	0	1	0	2	0	2	1
3	0.0740741	-0.5	0.769231	0	1	1	0	2	0	2	1
4	0.148148	-0.5	0.307692	-0.125	0	0	1	2	0	2	1

Validating outliers using boxplot



Now, We need to address these outliers



Outlier removal using Z Score & Winsorize method - trtbps variable

Process :

1. Import scipy and use zscore and winsorize.
2. Determine the threshold
3. Get the zscore for all the values and filter out the zscore > threshold
4. Find the min of the value in zscore list and corresponding max value from the Dataframe
5. Get the probability percentile of the trtbps column with respect to the min value
6. Use Winsorize function to filterout the values from the trtbps column > store it new variable
7. Now visualize the new variable using box plot
8. There will be no outliers

```
Threshold Value : 1
Number of Outliers : 51
insexex of Outliers : [  8   9  14  17  18  23  25  26  37  38  39  83 101 106 110 111 127 130
 145 147 150 152 165 179 181 184 195 196 202 203 204 207 220 223 228 231
 232 240 241 243 247 248 254 258 260 266 283 287 292 293 297]
------------------------------------------------

Threshold Value : 2
Number of Outliers : 13
insexex of Outliers : [  8 101 110 152 195 203 223 228 241 248 260 266 292]
------------------------------------------------
Threshold Value : 3
Number of Outliers : 2
insexex of Outliers : [223 248]
------------------------------------------------
```

Normal Distribution

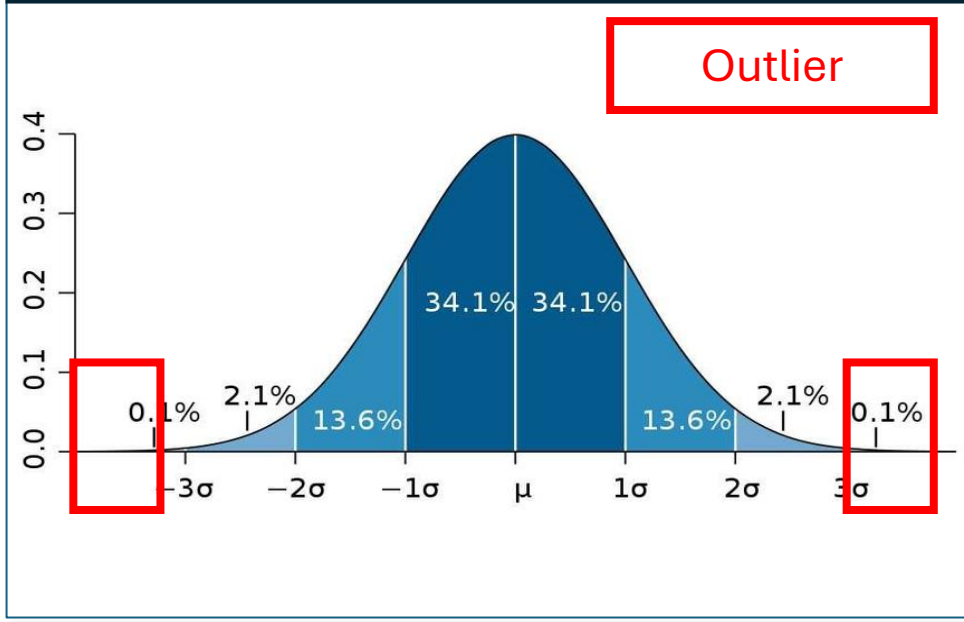
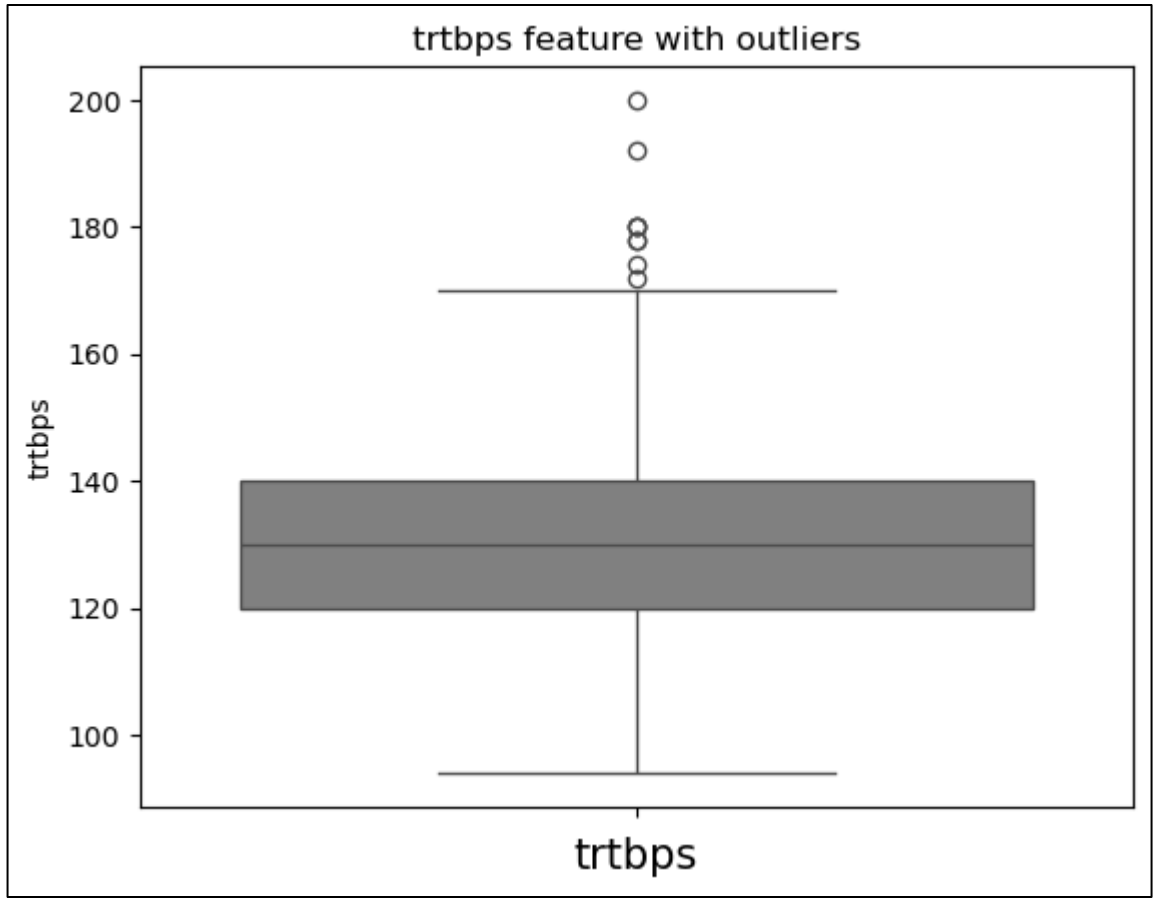


Image Source :
https://news.mit.edu/2012/explained-sigma-0209

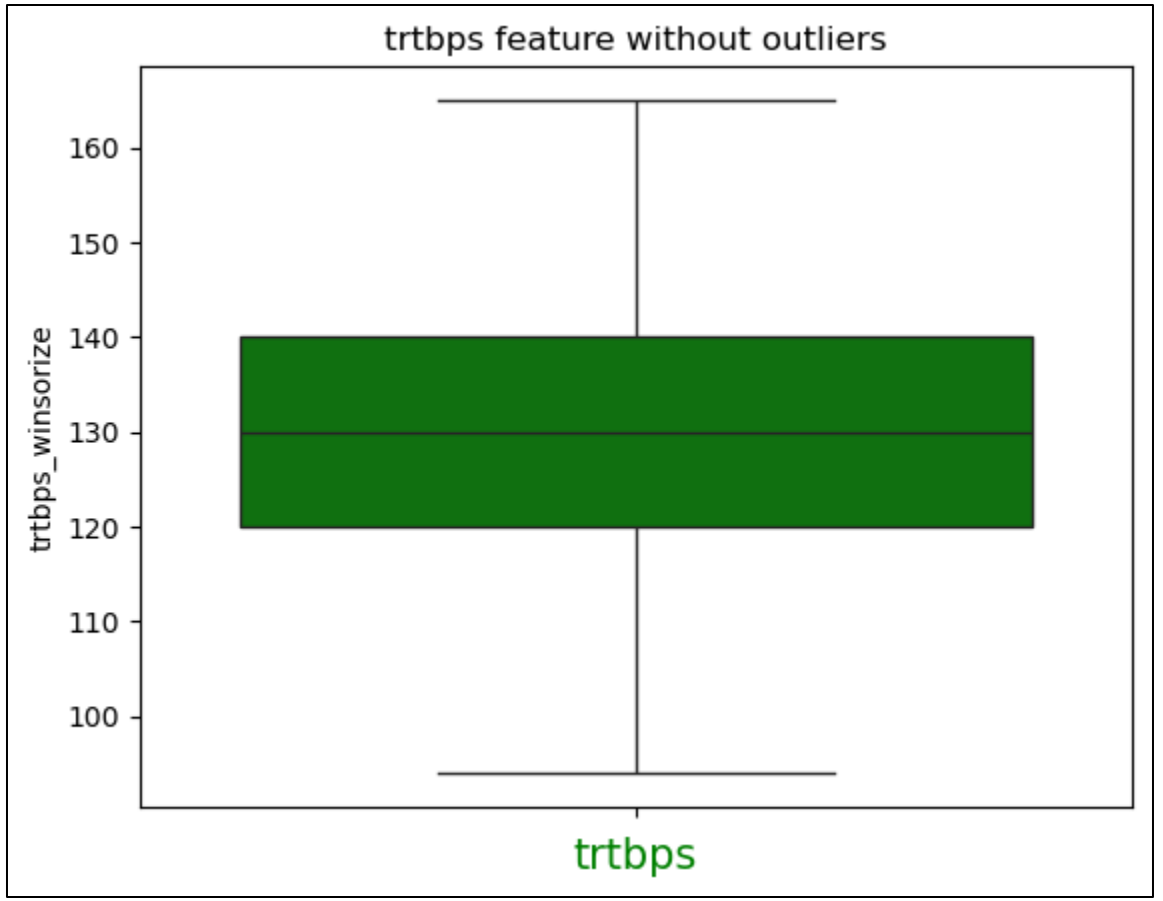
A Z-score measures how many standard deviations a raw data point is from the mean of a population

Outlier removal using Z Score - trtbps variable

Before outlier treatment



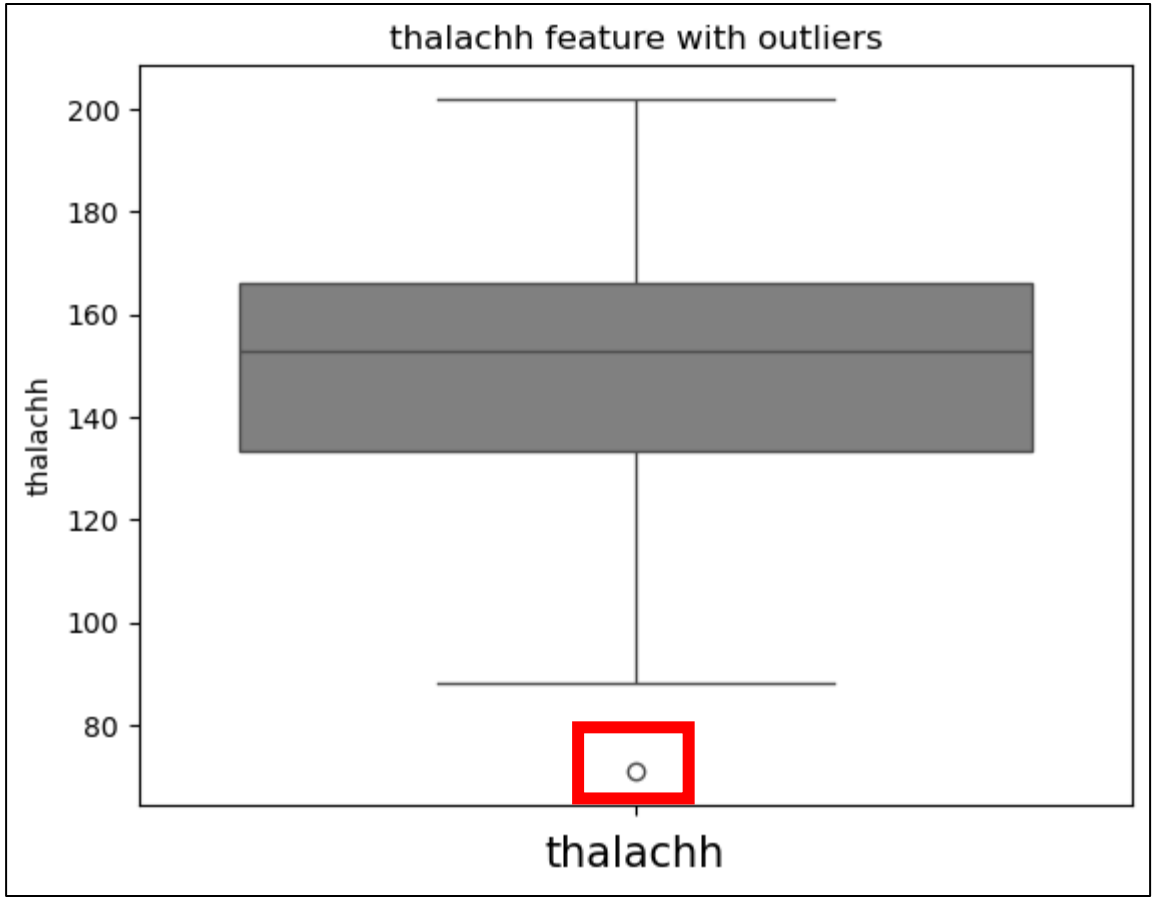
After outlier treatment



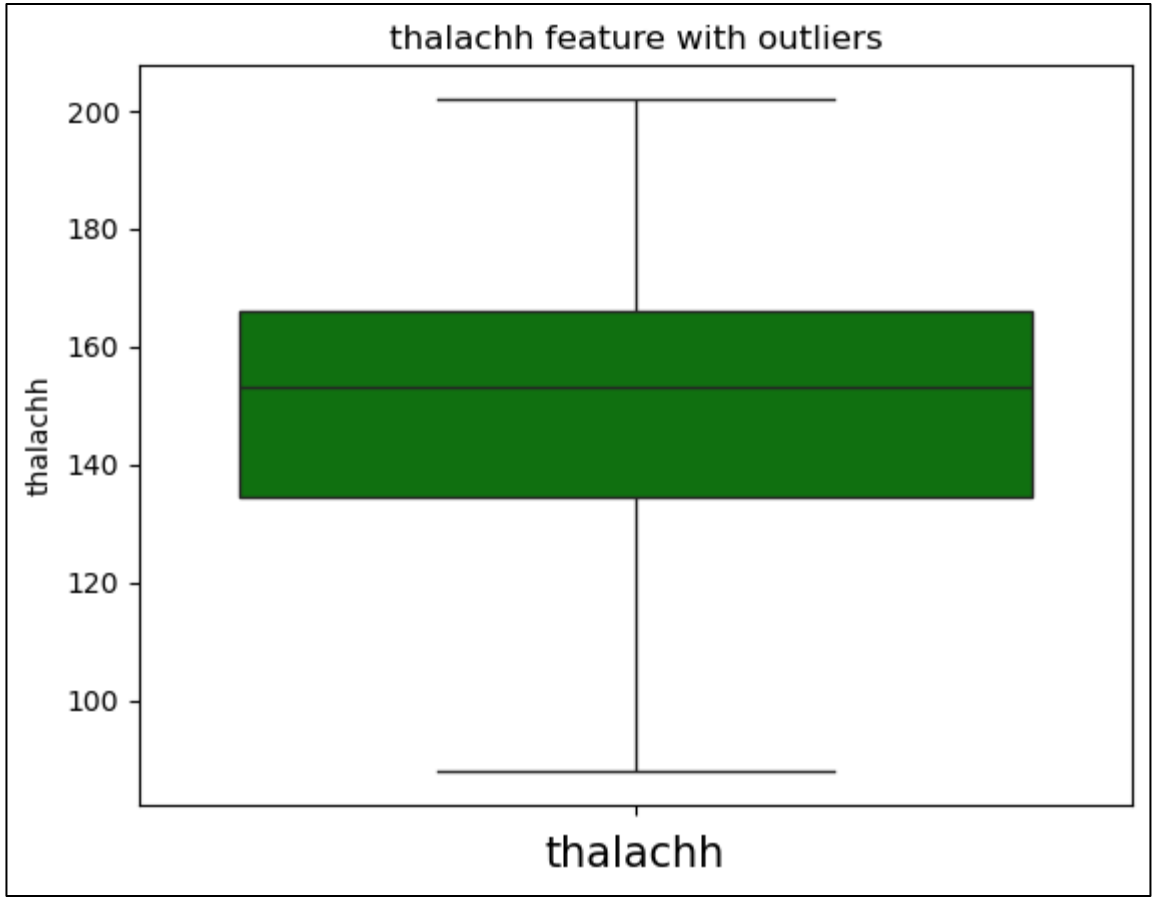
Outlier removal - thalachh variable – IQR Treatment

Outlier

Before outlier treatment



After outlier treatment

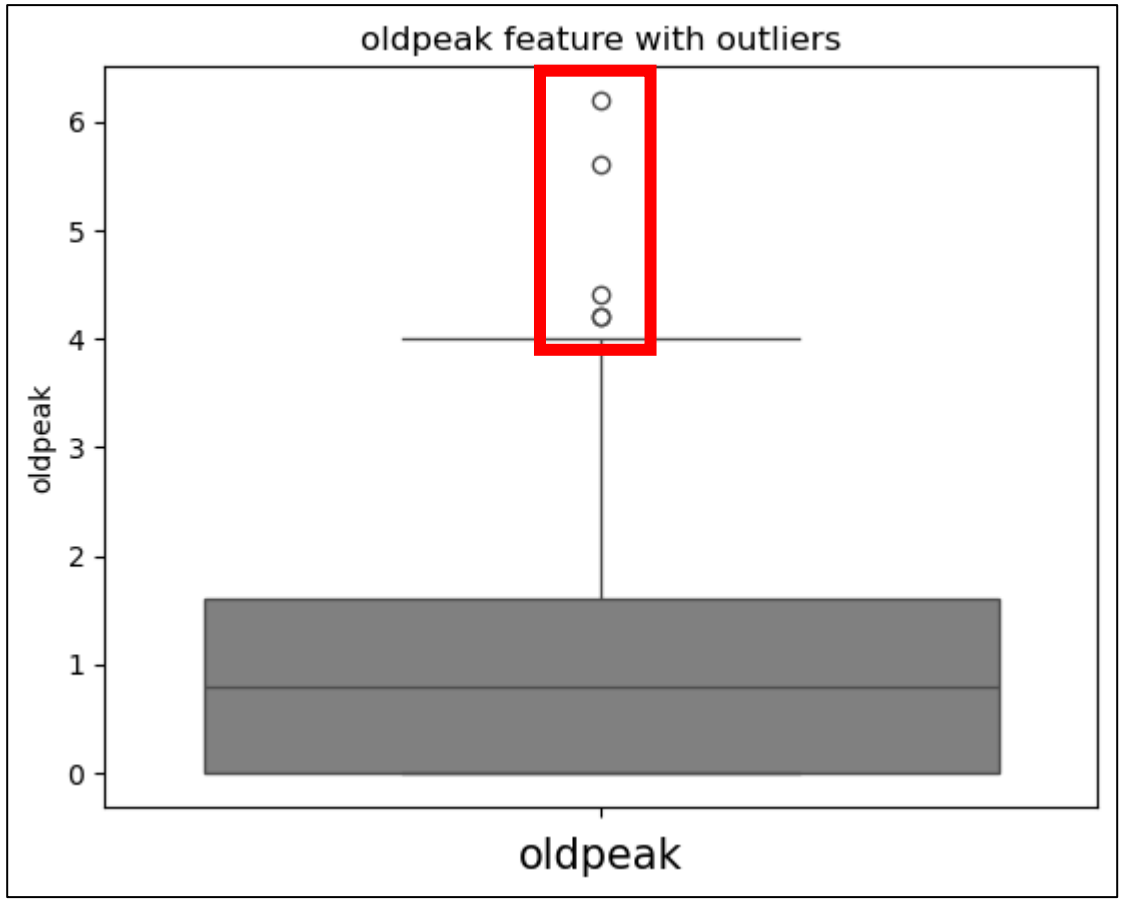


	age	sex	cp	trtbps	chol	fbs	restecg	thalachh	exng	oldpeak	slp	caa	thall	output
272	67	1	0	120	237	0	1	71	0	1	1	0	2	0

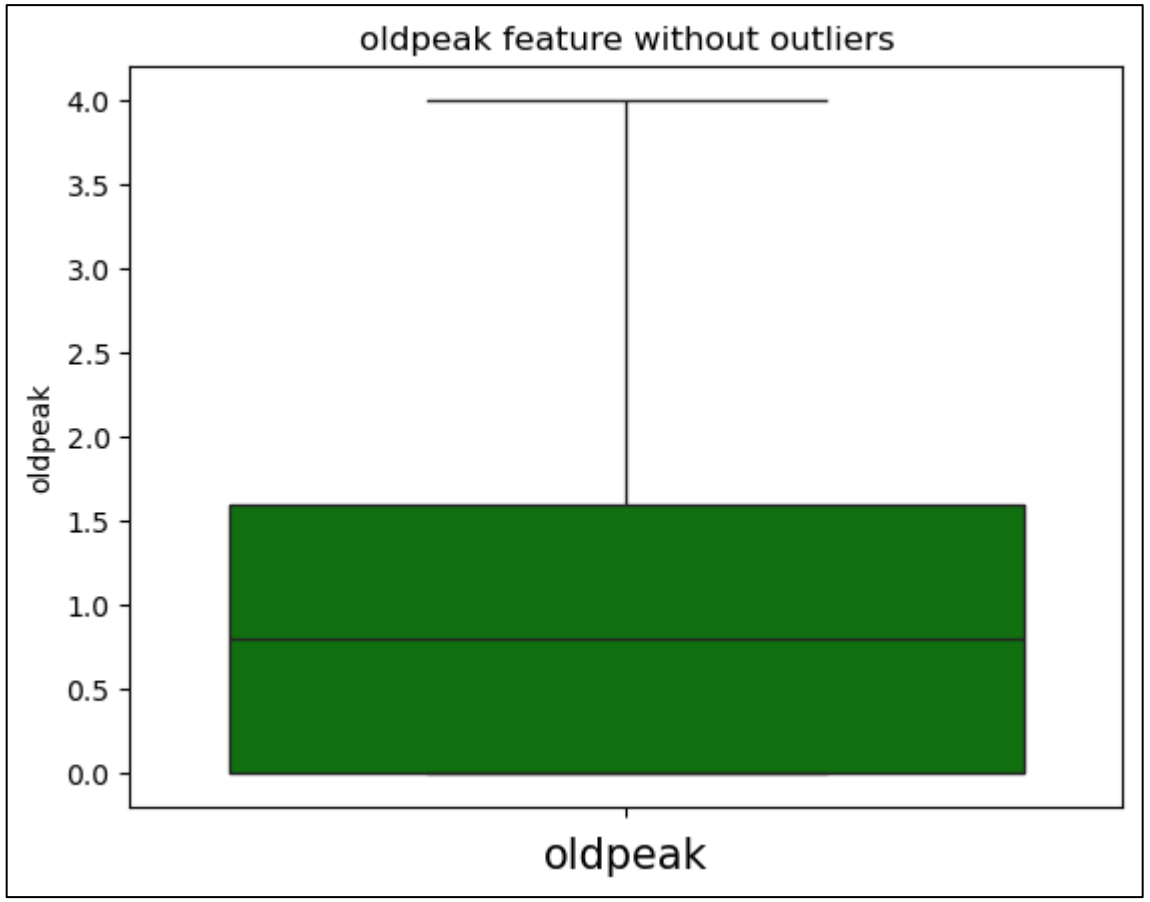
Outlier treatment - oldpeak variable – IQR & Winsorize

Outlier

Before outlier treatment



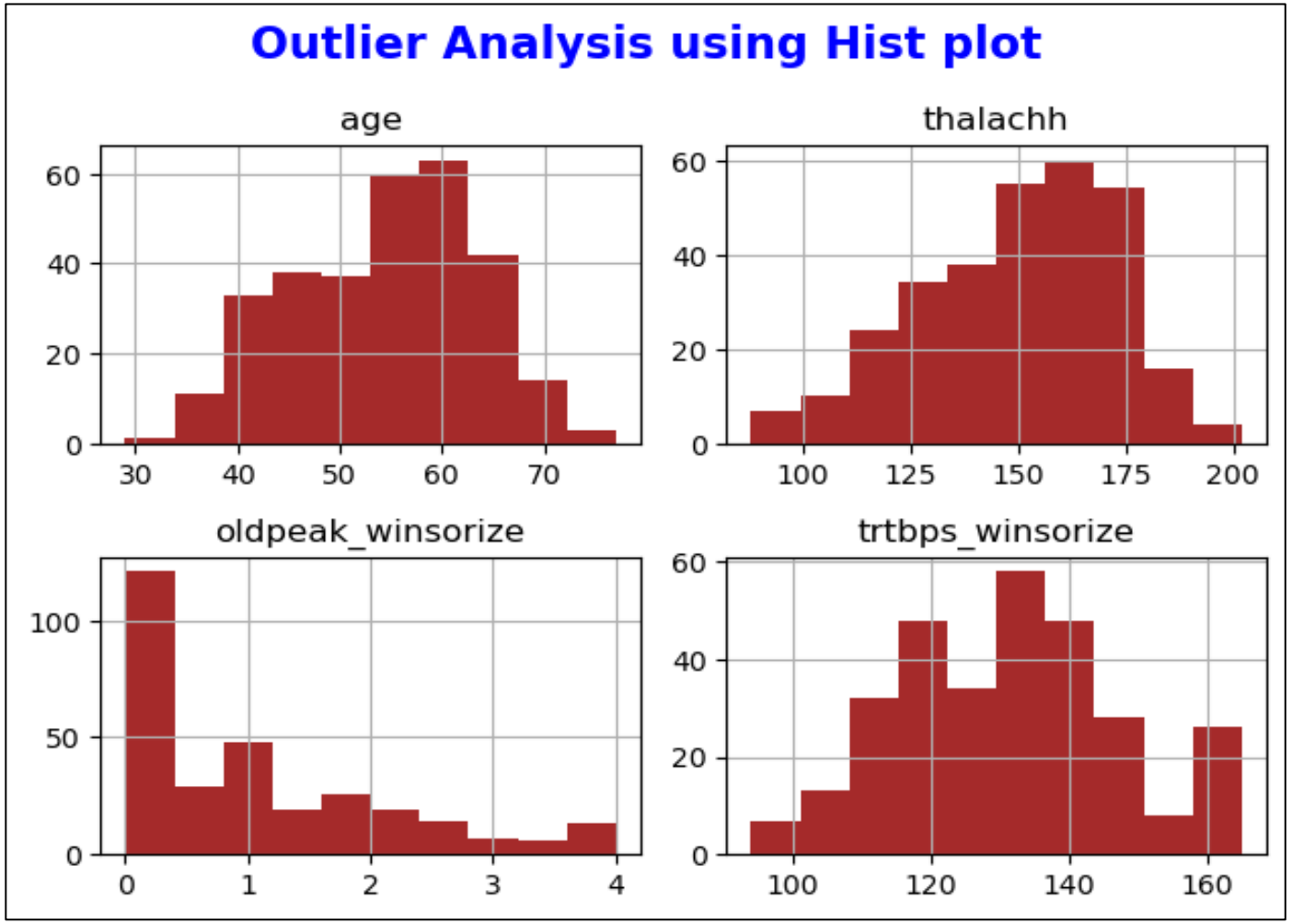
After outlier treatment



Validating Skewness of numerical variables

Skewness Meaning :
−0.5 to +0.5: Approximately symmetrical.
−1 to −0.5 or +0.5 to +1: Slightly skewed.
Less than −1 or greater than +1: Moderately or substantially skewed. Values beyond −2 and +2 are often indicative of significant non-normality.

	skew
age	-0.1992
thalachh	-0.4616
oldpeak_winsorize	0.9960
trtbps_winsorize	0.2520



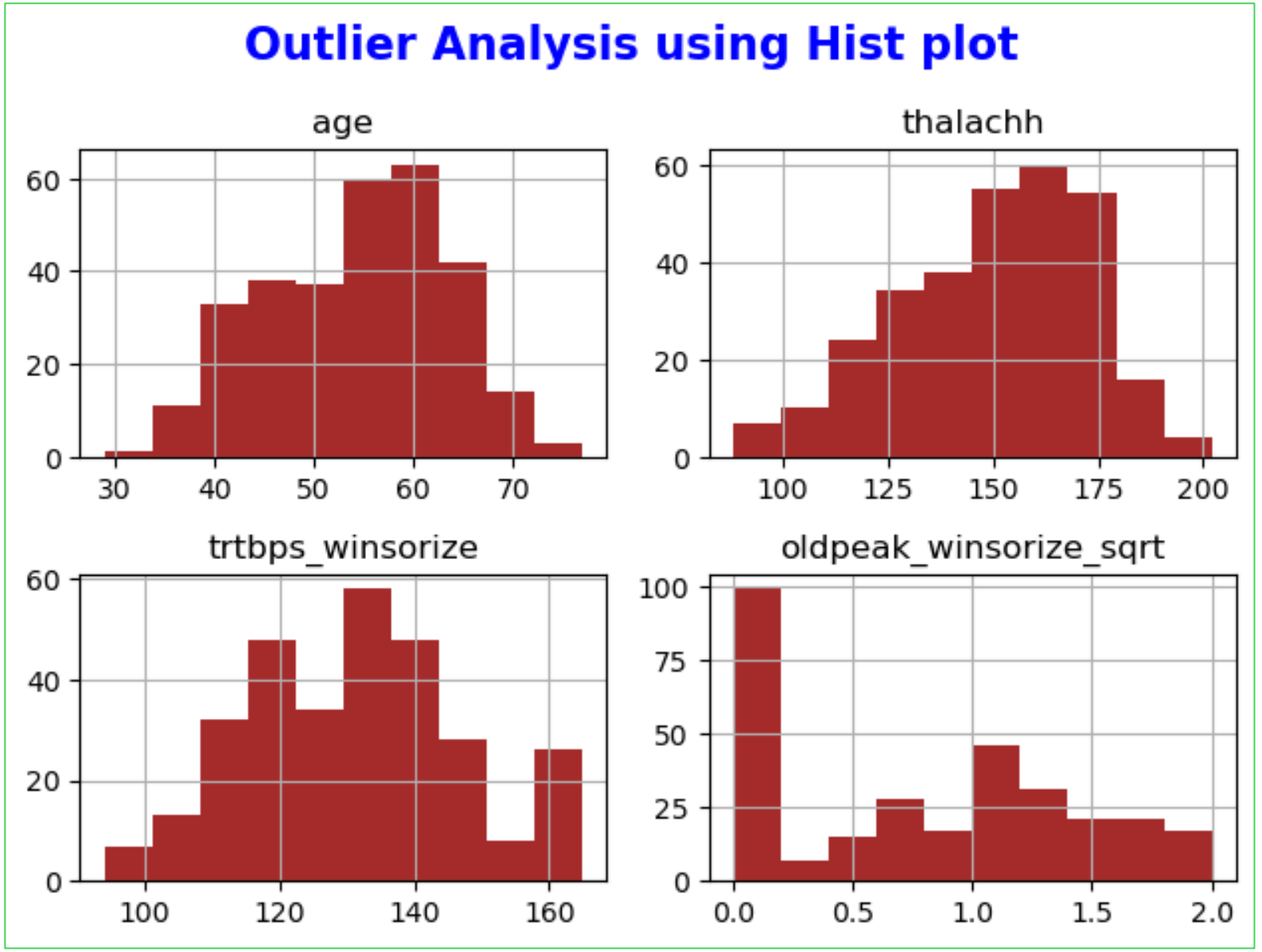
Inference > we need to reduce the skewness on oldpeak

Decreasing skewness in oldpeak variable

Two methods used:
- Logarithmic
- Square root

	skew
oldpeak_winsorize	0.996036
oldpeak_winsorize_log	NaN
oldpeak_winsorize_sqrt	0.108926

This skewness is close to what we want as **we see the data spread**



Categorical Variables – Treating with one hot encoding

sex_1	cp_1	cp_2	cp_3	exng_1	slp_1	slp_2	caa_1	caa_2	caa_3	caa_4	thall_2	thall_3
1	0	0	1	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0	1	0
0	1	0	0	0	0	1	0	0	0	0	1	0
1	1	0	0	0	0	1	0	0	0	0	1	0
0	0	0	0	1	0	1	0	0	0	0	1	0

Categorical variables need to be one hot encoded[0/1] for machine learning process

Scaling the numerical features using Robust Scalar method

	age	thalachh	trtbps_winsorize	oldpeak_winsorize_sqrt
0	63	150	145	1.51658
1	37	187	130	1.87083
2	41	172	130	1.18322
3	56	178	120	0.894427
4	57	163	120	0.774597

```
from sklearn.preprocessing import RobustScaler

robust_scaler_usdf = RobustScaler()
(Range : -1 to 1)
```

	age	thalachh	trtbps_winsorize	oldpeak_winsorize_sqrt
0	0.581818	-0.0952381	0.75	0.491851
1	-1.30909	1.07937	0	0.771913
2	-1.01818	0.603175	0	0.228308
3	0.0727273	0.793651	-0.5	0
4	0.145455	0.31746	-0.5	-0.0947343

Summary before machine learning process

Numerical Features

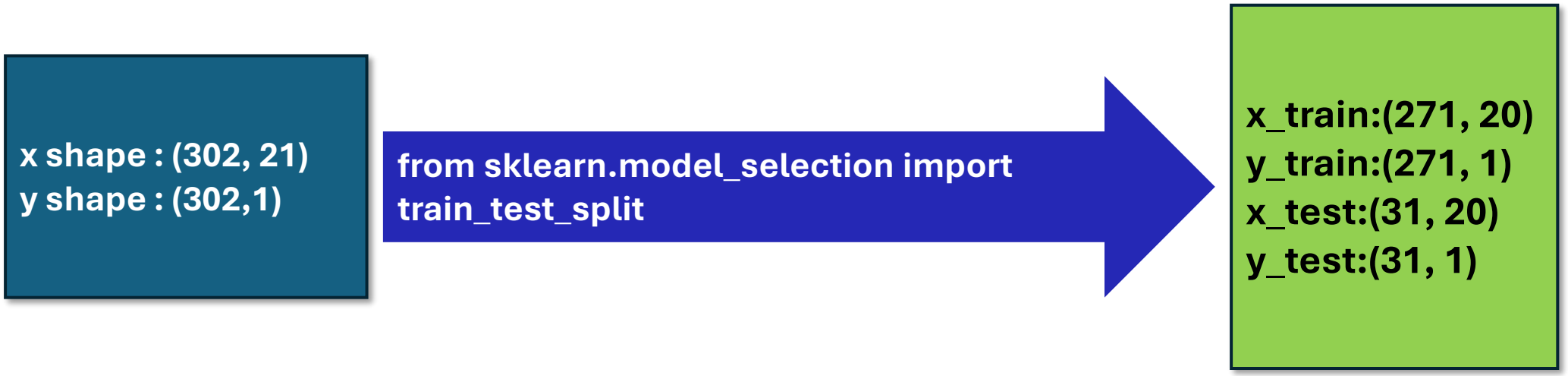
- Analyzed using boxplot, histplots and heatmaps.
- Treated for outliers with IQR, Zscore and winsorize method.
- Treated for skewness
- Finally Scaled using Robust Scaler method
- One numerical feature was removed due to low correlation

Categorical features

- Analyzed using boxplot, histplots and heatmaps
- Treated for outliers
- Finally, One hot encoding implemented
- Two features were removed due to low or no correlation.

Machine Learning Process

Splitting of Train and Test data using train_test_split



Machine learning using Logistic regression: Accuracy and Reliability

Logistic regression is used for classification objective and when dependent variable is categorical.

In our case we have the variable output

Model Reliability : Is the model memorizing ?
Checking for overfitting

accuracy_score
(y_true=y_test, y_pred=y_pred_usdf)

0.8709677419354839

Cross validation
accuracy_score for 10 folds
0.8484848484848484

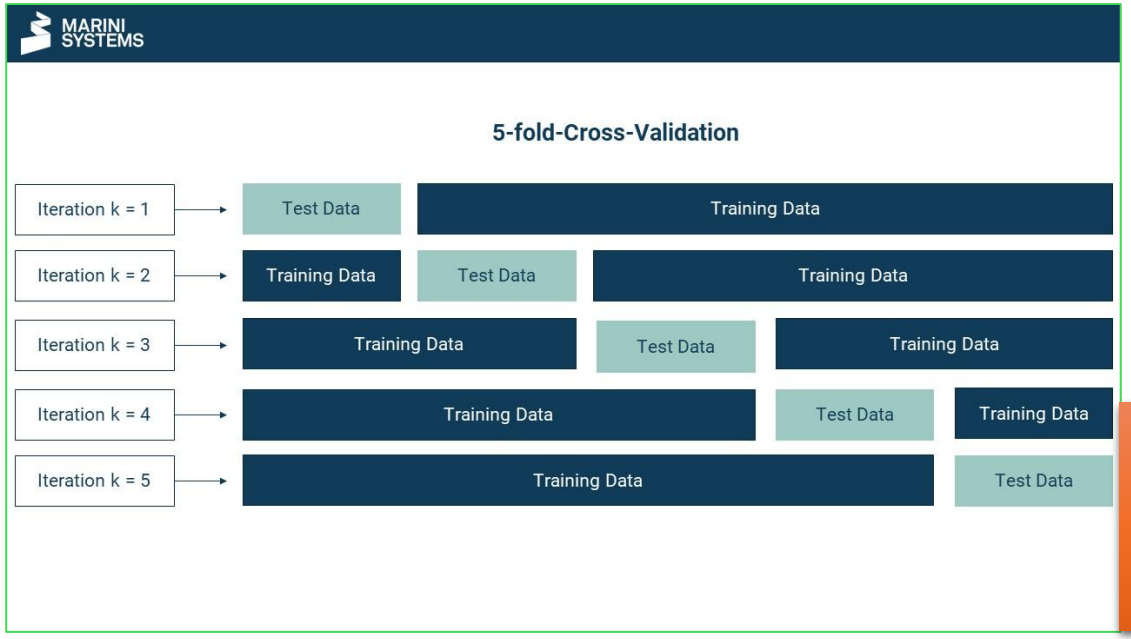


Image Source: https://marini.systems/en/glossary/cross-validation/

Inference:
No overfitting as cross validation scores is close to the predicted accuracy score.

Logistic regression : model performance - AUC

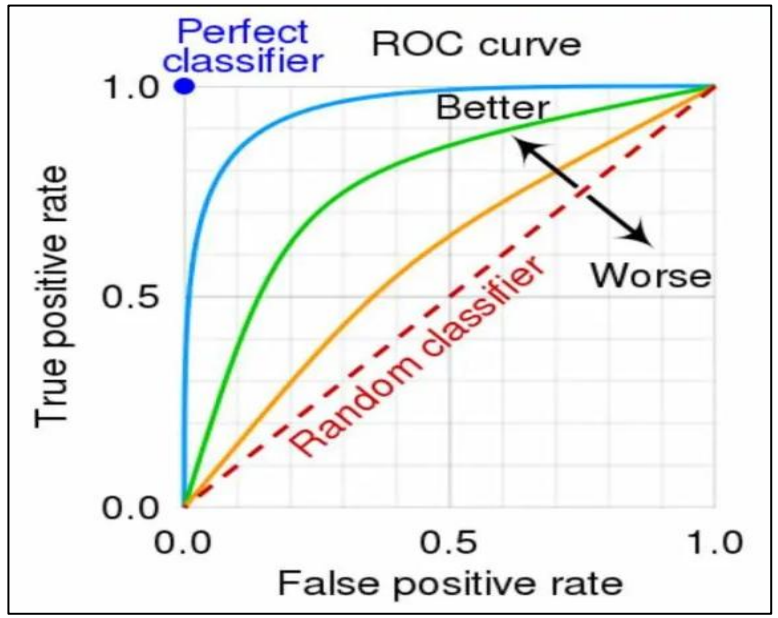
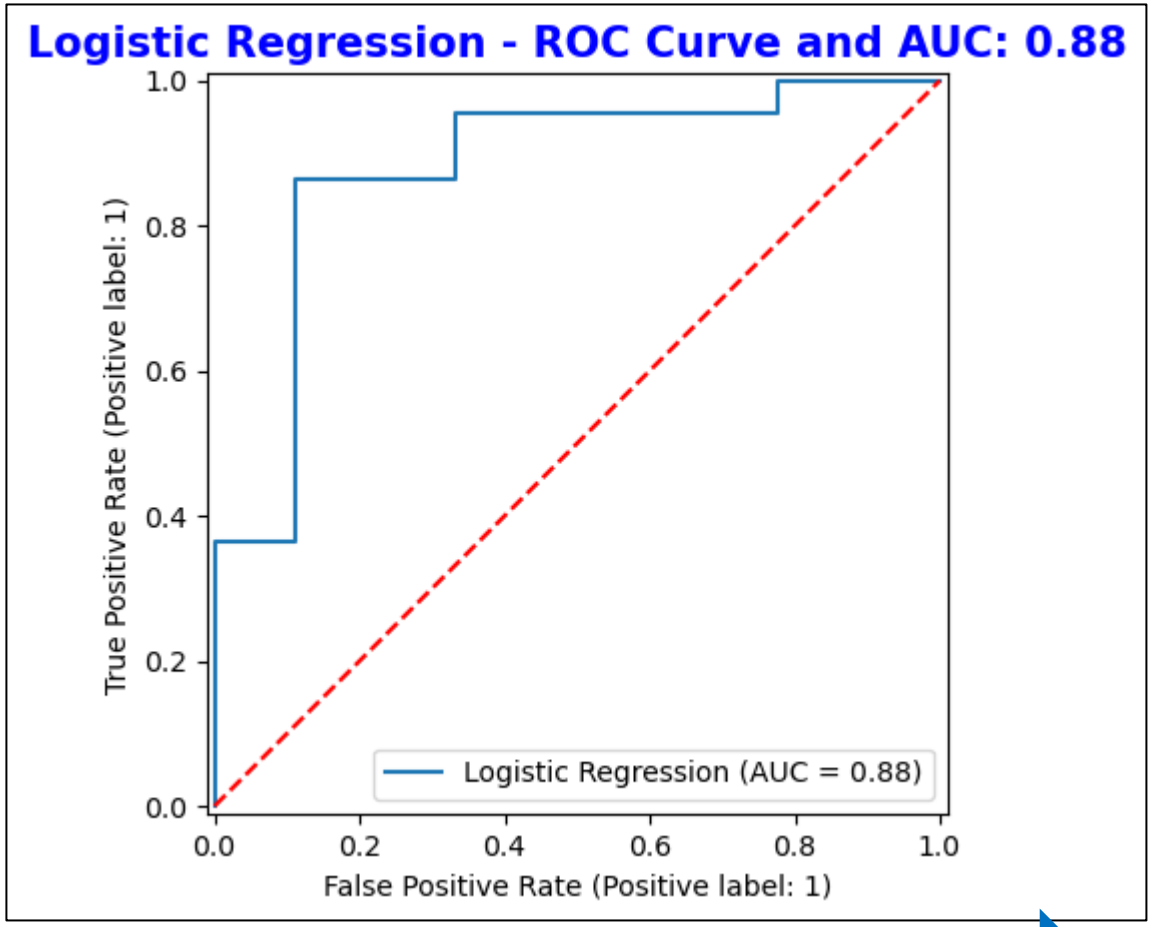


Image Source: https://spotintelligence.com/2024/06/17/roc-auc-curve-in-machine-learning/

Y axis : True Positive Rate = Recall sensitivity = letting us know the true predictability i.e. 0 as o and 1 as 1

Better Area under the curve – better model

AUC Values	Test Quality
0.9 - 1.0	Excellent
0.8 - 0.9	Very good
0.8 - 0.7	Good
0.7 - 0.6	Satisfactory
0.6 - 0.5	Unsatisfactory



Inference:
AUC is very good

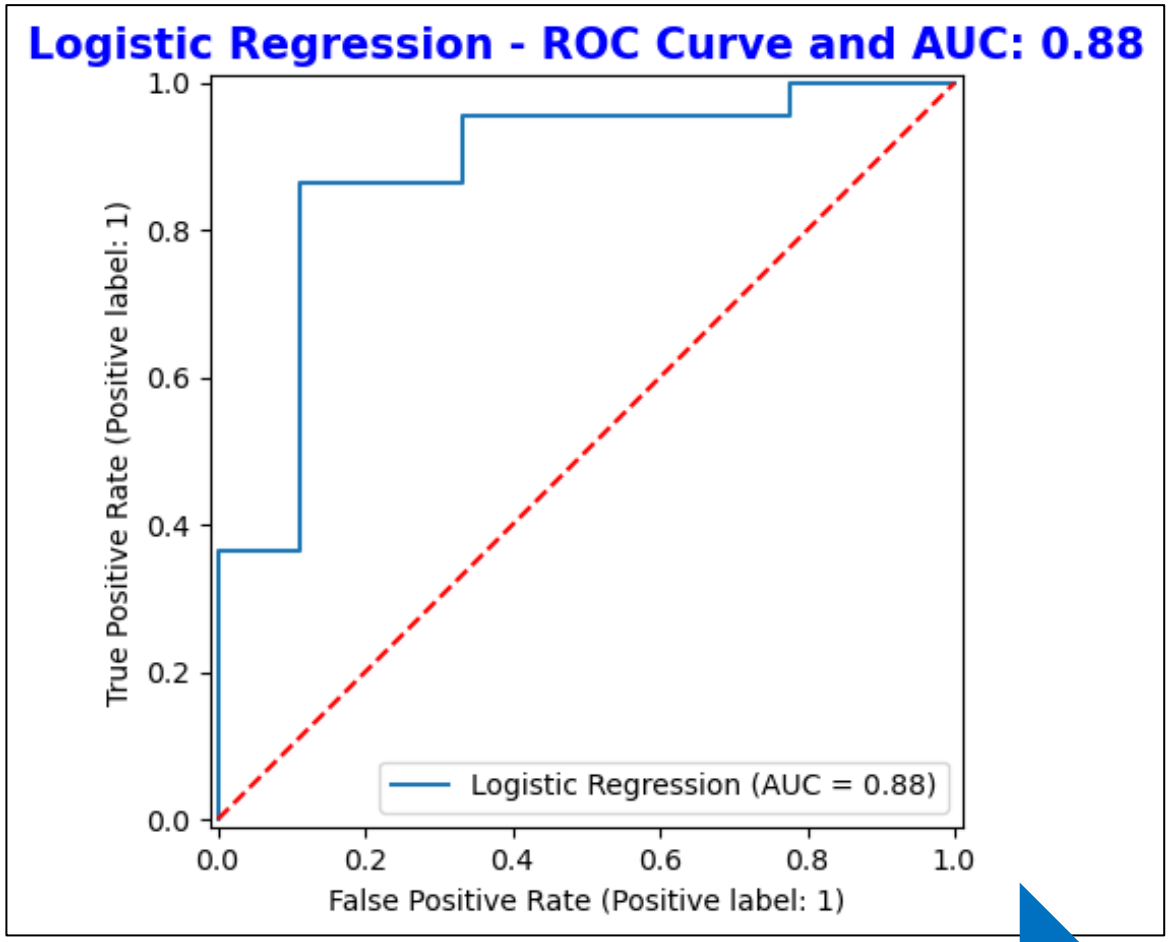
Logistic Regression : Hyperparameter Optimization using GridsearchCV

Hyper Parameter Optimization for success of our model with Grid Seach CV

```
1  from sklearn.model_selection import GridSearchCV
2
3  log_reg_ml_usdf_new = LogisticRegression()
4  log_reg_hyp_param_usdf = {'penalty': ['l1', 'l2', 'elasticnet'], 'solver': [
5      'lbfgs', 'liblinear', 'newton-cg', 'newton-cholesky', 'sag', 'saga']}
6  log_reg_grid = GridSearchCV(log_reg_ml_usdf_new, log_reg_hyp_param_usdf)
7  log_reg_grid.fit(x_train, y_train)
8
9  print(f'Logistic Regression Best Parameters : {log_reg_grid.best_params_}')
```

✓ 0.4s

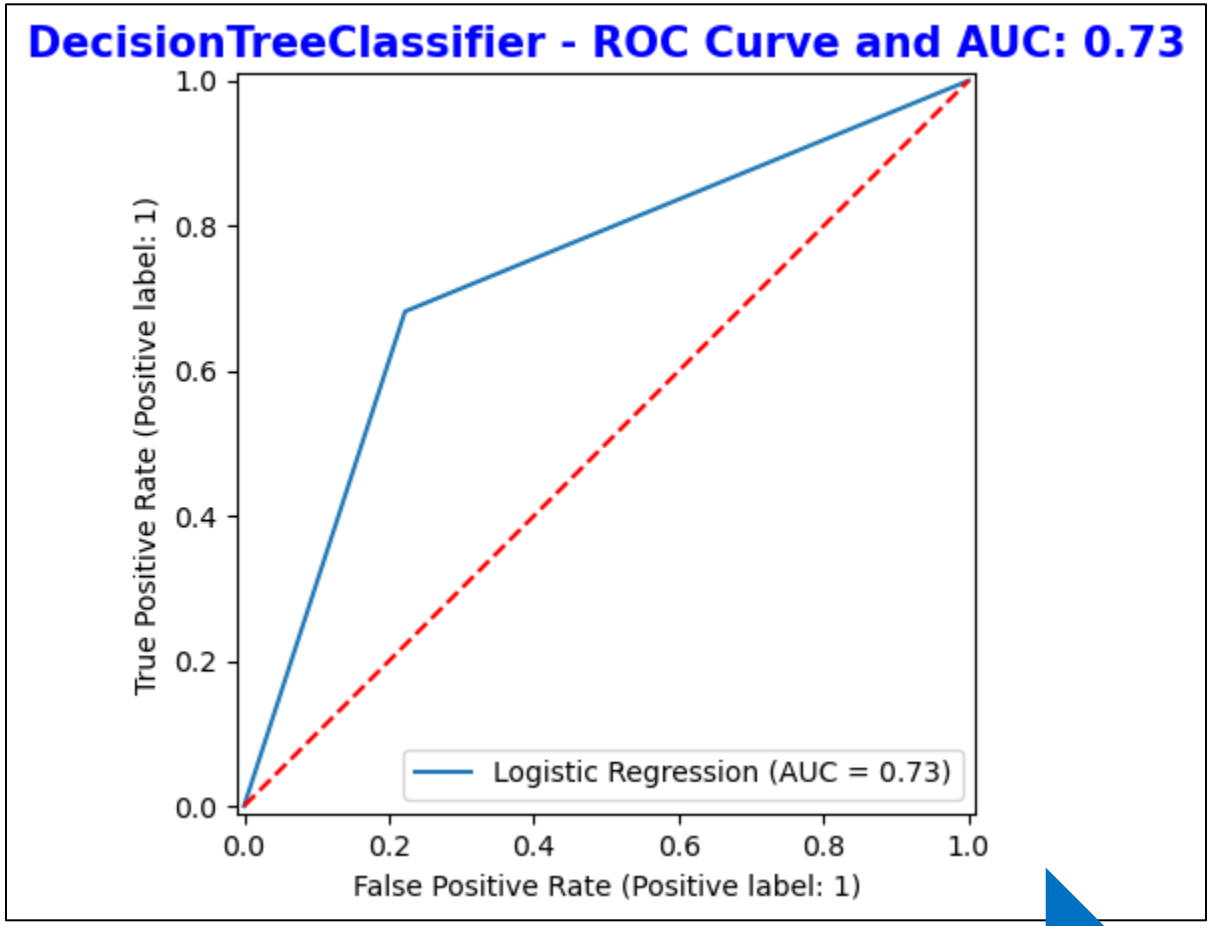
```
Logistic Regression Best Parameters : {'penalty': 'l2', 'solver': 'liblinear'}
```



Inference on AUC:
AUC is very good and similar after hyper Param optimization

Decision Tree Classifier

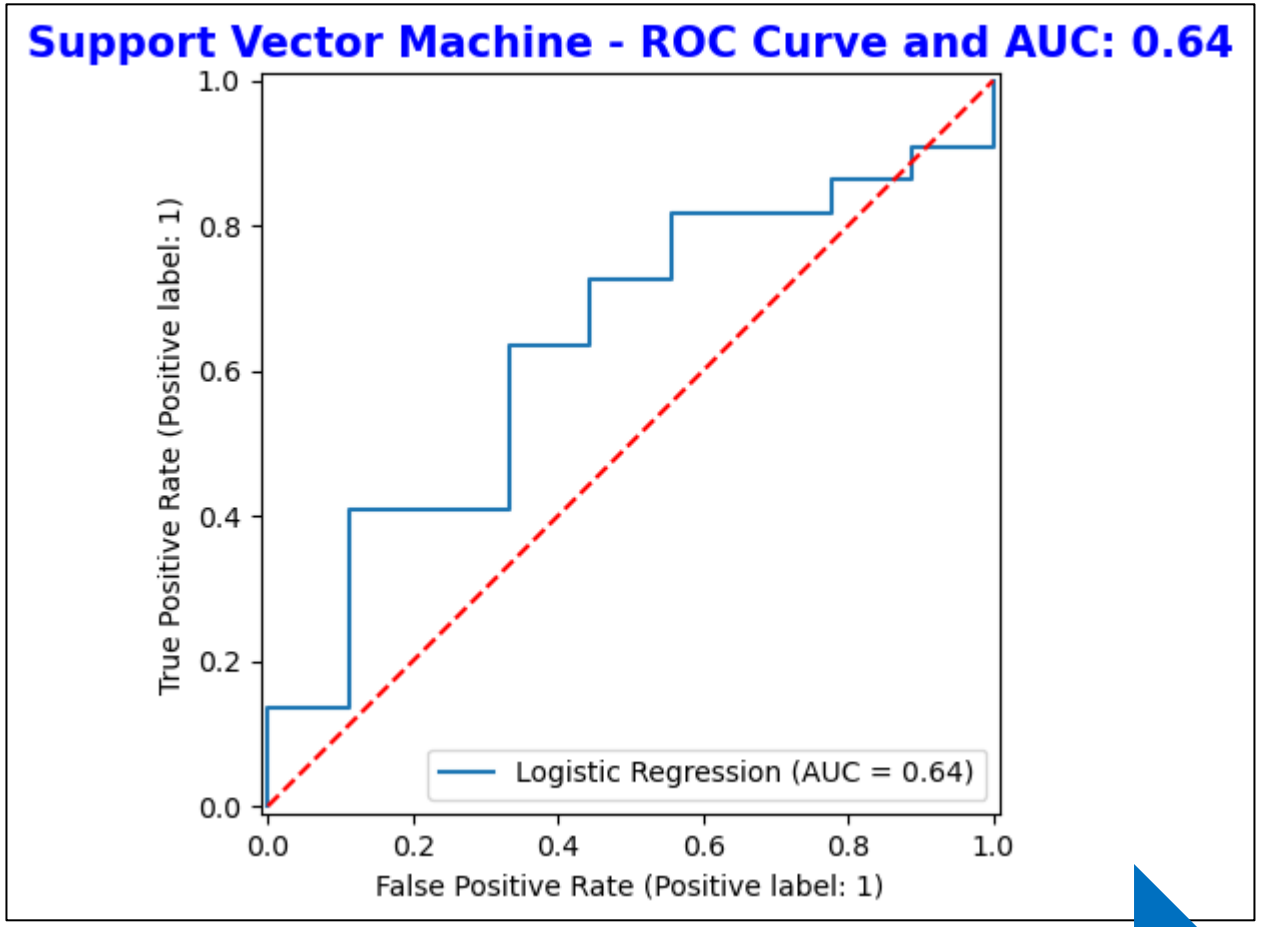
Desicion tree accuracy_score :
0.7096774193548387
Cross Validation Accuracy Scores Average:
0.8484848484848484



Inference on AUC:
Logistic Regression Classifier is better than **Decision Tree AUC is less**

SVM classifier

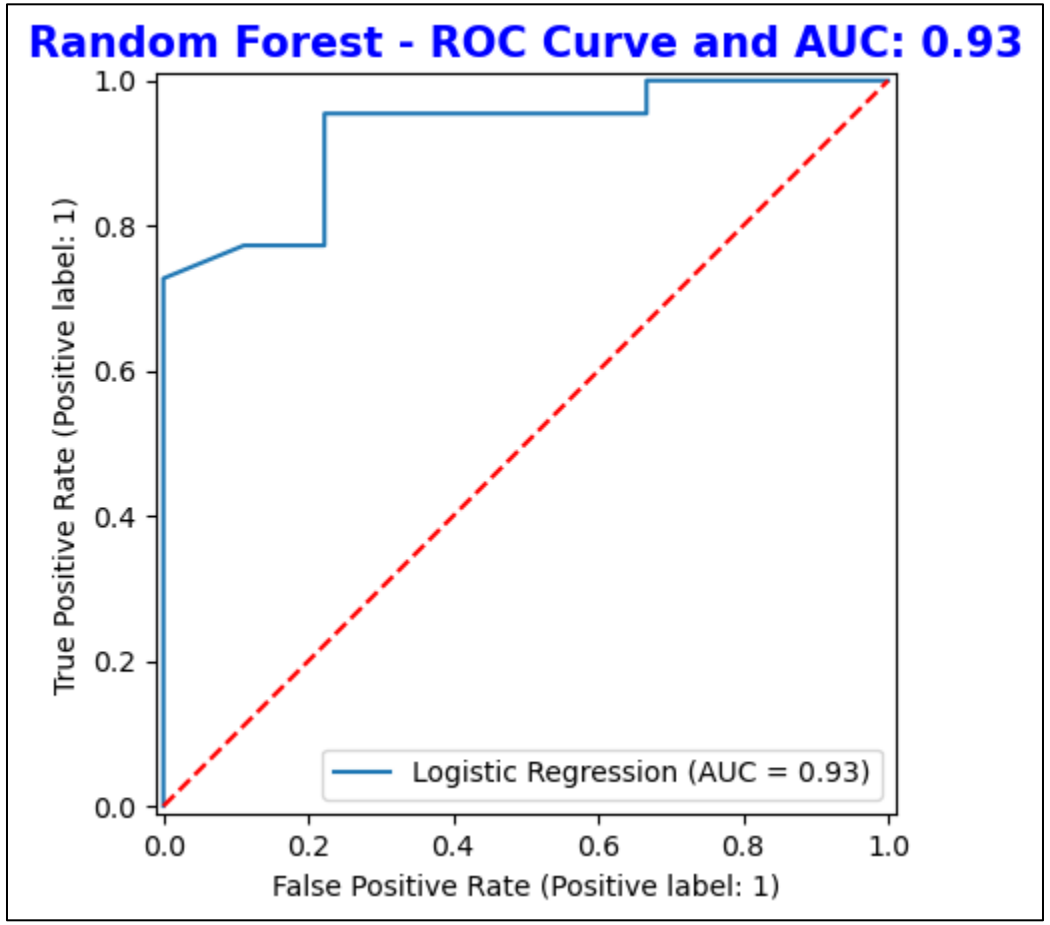
SVM accuracy_score :
0.6451612903225806
Cross Validation Accuracy Scores Average:
0.7272727272727273



Inference on AUC:
Logistic Regression Classifier is better than **SVM Classifier**

Random Forest classifier

Random Forest accuracy_score :
0.6451612903225806
Cross Validation Accuracy Scores Average:
0.9090909090909091



Inference on AUC:
Random Forest Classifier is better than Logistic Regression

Summary of Model scores

Model	Crossvalidation Accuracy	AUC value
Linear Regression	0.87	0.88
Decision Tree	0.85	0.73
Support Vector Machine	0.73	0.64
Random Forest Classifier	0.91	0.93

Summary of Models

Models tested:		Result

Models tested:
Linear Regression
Decision Tree
Support Vector Machine
Random Forest Classifier

Machine Learning Study Process

Model	Crossvalidation Accuracy	AUC value
Linear Regression	0.87	0.88
Decision Tree	0.85	0.73
Support Vector Machine	0.73	0.64
Random Forest Classifier	0.91	0.93

Good Cross Validation Accuracy:
Random Forest Classifier

Good AUC :
Random Forest Classifier

Thankyou